Modelling Exponential Growth and Decay

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Exponentials and Logarithms (ref `el5`)

This document contains all 70 *Modelling Exponential Growth and Decay* questions, each with a fully worked solution and a TikZ graph of the model. Each diagram plots the model against time with the initial value and any evaluated/target points marked, the long-term asymptote shown dashed for bounded (cooling, logistic, capped-growth) models, both curves with their crossover for comparison questions, and a tangent for the rate-of-change questions.

Contents

Modelling Exponential Growth and Decay 01 (el5-001)

Difficulty: Foundation | *Marks: 3*

A colony of bacteria is modelled by $B = 200e^{0.05t}$, where B is the number of bacteria and t is the time in hours.

(a) Write down the initial number of bacteria.

(b) Predict the number of bacteria after 10 hours. Give your answer to the nearest whole number.

Worked solution.

Step 1: (a) Substitute $t = 0$.

$$B = 200e^{0.05\times 0} = 200e^0 = 200$$

$e^0 = 1$, so the initial number of bacteria is 200.

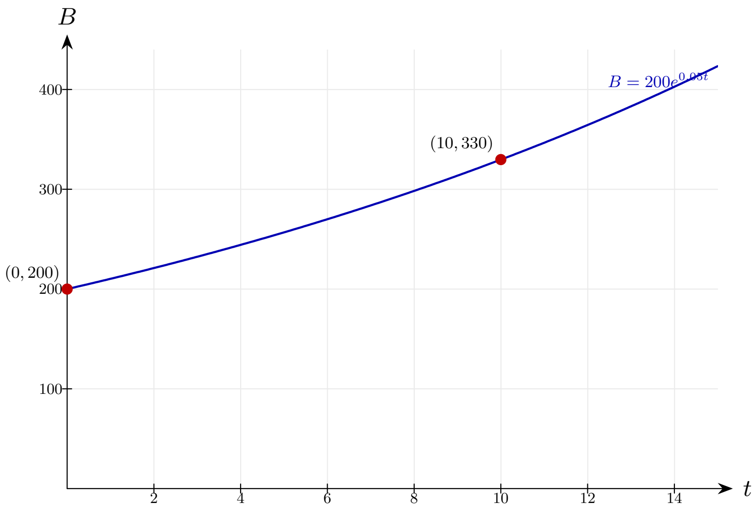
Step 2: (b) Substitute $t = 10$.

$$B = 200e^{0.05\times 10} = 200e^{0.5} \approx 200 \times 1.6487 \approx 330$$

Use a calculator to evaluate $e^{0.5}$. Round to the nearest whole number.

Final answer: (a) 200 bacteria (b) 330 bacteria

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 02 `(e15-002)`

Difficulty: Foundation | *Marks: 3*

The mass M (grams) of a radioactive sample after t years is given by $M = 500e^{-0.03t}$.
(a) State the initial mass.
(b) Find the mass remaining after 20 years. Give your answer to 3 significant figures.

Worked solution.

Step 1: (a) Substitute $t = 0$.

$$M = 500e^0 = 500 \text{ grams}$$

The initial mass is 500 g.

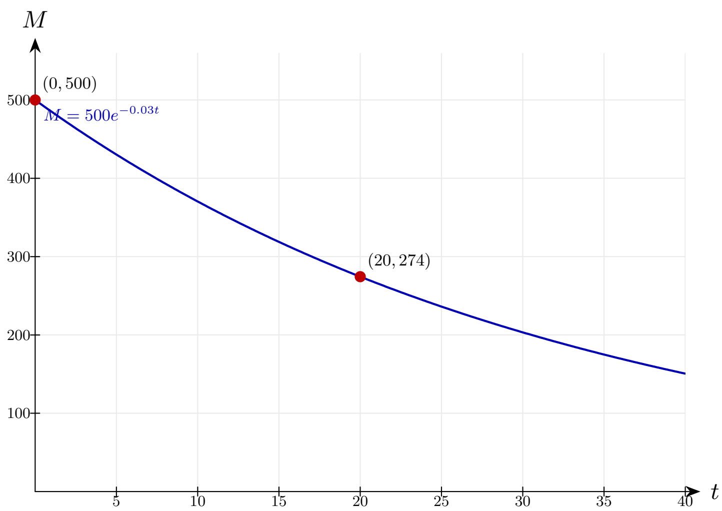
Step 2: (b) Substitute $t = 20$.

$$M = 500e^{-0.03\times 20} = 500e^{-0.6} \approx 500 \times 0.5488 \approx 274 \text{ g}$$

Use a calculator to evaluate $e^{-0.6}$.

Final answer: (a) 500 g (b) 274 g

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 03 (el5-003)

Difficulty: Foundation | Marks: 3

An oven is preheating. Its temperature T ℃ after t minutes is given by $T = 220 - 200e^{-0.1t}$.

(a) What was the initial temperature of the oven?

(b) What is the temperature after 8 minutes? Give your answer to 3 s.f.

Worked solution.

Step 1: (a) Substitute $t = 0$.

$$T = 220 - 200e^{0} = 220 - 200 = 20\,℃$$

The oven starts at room temperature, 20°C.

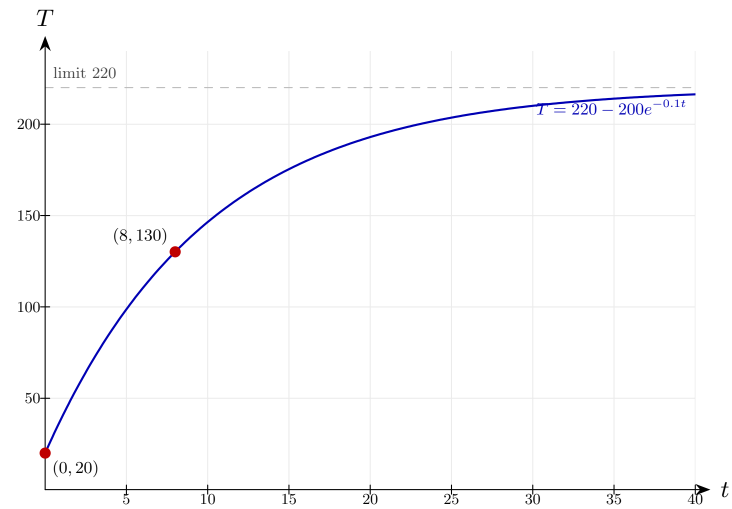
Step 2: (b) Substitute $t = 8$.

$$T = 220 - 200e^{-0.8} \approx 220 - 200 \times 0.4493 \approx 220 - 89.9 \approx 130\,℃$$

Evaluate $e^{-0.8} \approx 0.4493$ using a calculator.

Final answer: (a) 20 ℃ (b) 130 ℃

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 04 (el5-004)

Difficulty: Foundation | Marks: 3

The value V (£) of a motorbike t years after purchase is modelled by $V = 8000e^{-0.15t}$.

(a) State the purchase price.

(b) Find the value after 4 years. Give your answer to the nearest pound.

Worked solution.

Step 1: (a) Substitute $t = 0$.

$$V = 8000e^{0} = £8000$$

The initial purchase price is £8000.

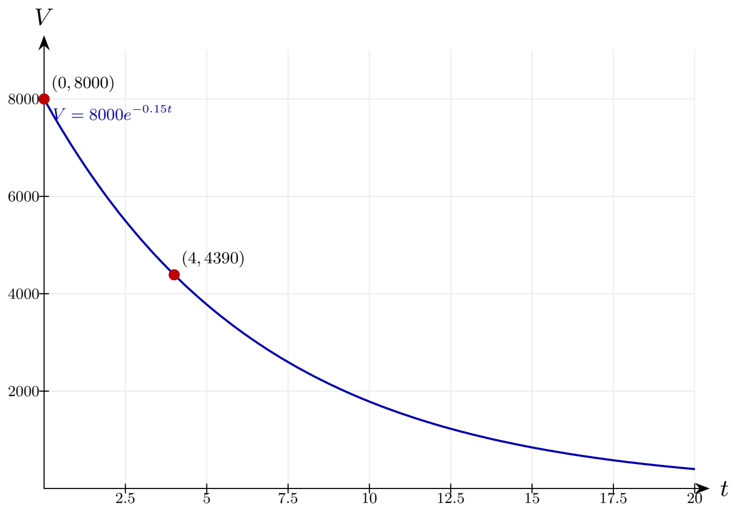
Step 2: (b) Substitute $t = 4$.

$$V = 8000e^{-0.6} \approx 8000 \times 0.5488 \approx £4390$$

Evaluate $e^{-0.6} \approx 0.5488$.

Final answer: (a) £8000 (b) £4390

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 05 `(e15-005)`

Difficulty: Foundation | Marks: 3

A savings account earns compound interest. The amount A **(£) after** t **years is modelled by** $A = 2500 \times 1.04^{t}$**.**

(a) Write down the initial amount invested.

(b) Find the amount in the account after 6 years. Give your answer to the nearest penny.

Worked solution.

Step 1: (a) Substitute $t = 0$.

$$A = 2500 \times 1.04^{0} = 2500 \times 1 = £2500$$

Any number to the power 0 is 1.

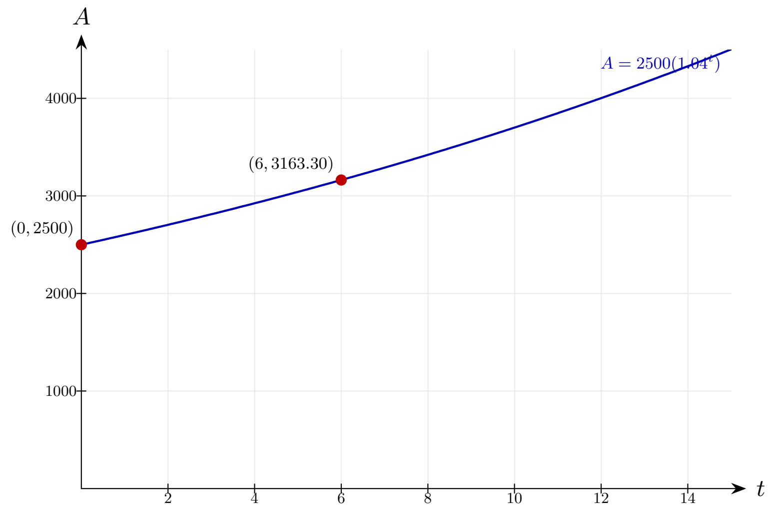
Step 2: (b) Substitute $t = 6$.

$$A = 2500 \times 1.04^{6} \approx 2500 \times 1.26532 \approx £3163.30$$

Evaluate $1.04^{6} \approx 1.26532$ using a calculator. Each factor of 1.04 represents one year of 4% growth, so after six years the amount has multiplied by roughly 1.265.

Final answer: (a) £2500 (b) £3163.30

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 06 `(e15-006)`

Difficulty: Foundation | *Marks: 3*

A drug concentration C (mg/litre) in a patient's bloodstream after t hours is modelled by $C = 4e^{-0.2t}$.

(a) State the initial concentration.

(b) Find the concentration after 3 hours. Give your answer to 3 s.f.

Worked solution.

Step 1: (a) Substitute $t = 0$.

$$C = 4e^0 = 4 \text{ mg/litre}$$

The initial concentration is 4 mg per litre of blood.

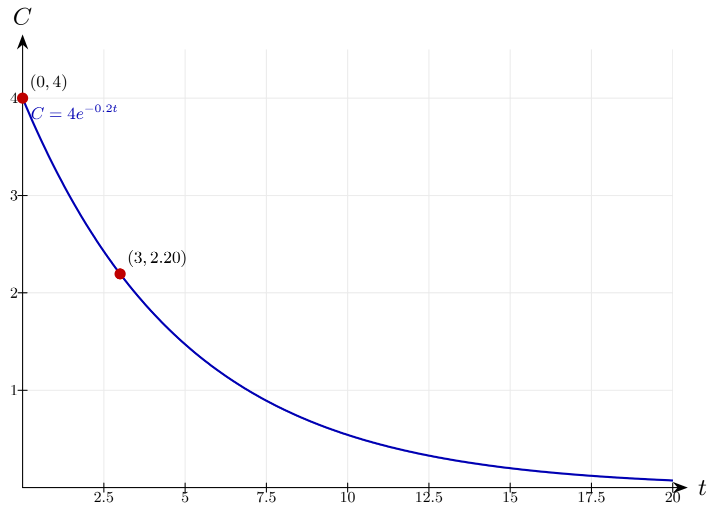
Step 2: (b) Substitute $t = 3$.

$$C = 4e^{-0.6} \approx 4 \times 0.5488 \approx 2.20 \text{ mg/litre}$$

Evaluate $e^{-0.6} \approx 0.5488$.

Final answer: (a) 4 mg/litre (b) 2.20 mg/litre

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 07 `(e15-007)`

Difficulty: Foundation | Marks: 4

A population of insects is modelled by $P = 150e^{0.06t}$, where t is the time in days. Find the value of t when the population first reaches 500. Give your answer to 3 significant figures.

Worked solution.

Step 1: Set $P = 500$ and solve.

$$150e^{0.06t} = 500 \Rightarrow e^{0.06t} = \frac{500}{150} = \frac{10}{3}$$

Divide both sides by 150.

Step 2: Apply ln to both sides.

$$0.06t = \ln\left(\frac{10}{3}\right)$$

$\ln(e^{0.06t}) = 0.06t$.

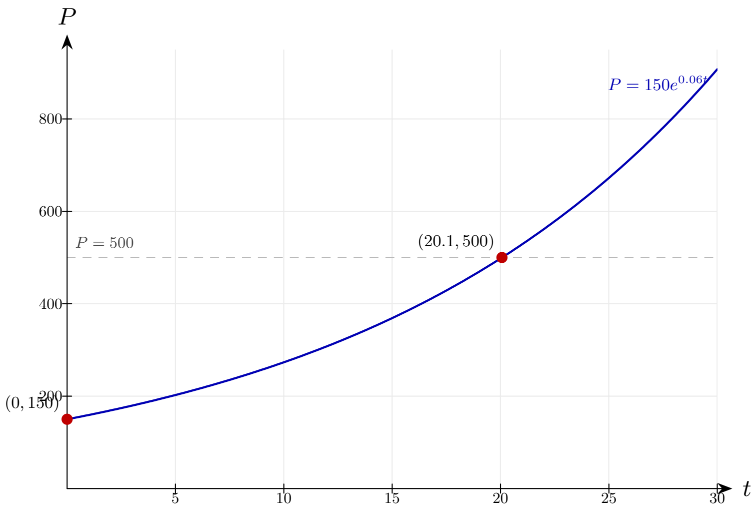
Step 3: Solve for t.

$$t = \frac{\ln(10/3)}{0.06} \approx \frac{1.2040}{0.06} \approx 20.1 \text{ days}$$

Evaluate using a calculator.

Final answer: $t \approx 20.1$ days

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 08 (e15-008)

Difficulty: Foundation | *Marks: 4*

The mass M grams of a substance decays according to $M = 800e^{-0.04t}$, where t is in years. Find the time taken for the mass to reduce to 200 g. Give your answer to 3 s.f.

Worked solution.

Step 1: Set $M = 200$.

$$800e^{-0.04t} = 200 \Rightarrow e^{-0.04t} = \frac{200}{800} = 0.25$$

Divide both sides by 800.

Step 2: Apply ln.

$$-0.04t = \ln(0.25) = -\ln 4$$

$0.25 = \frac{1}{4}$, so $\ln(0.25) = -\ln 4$.

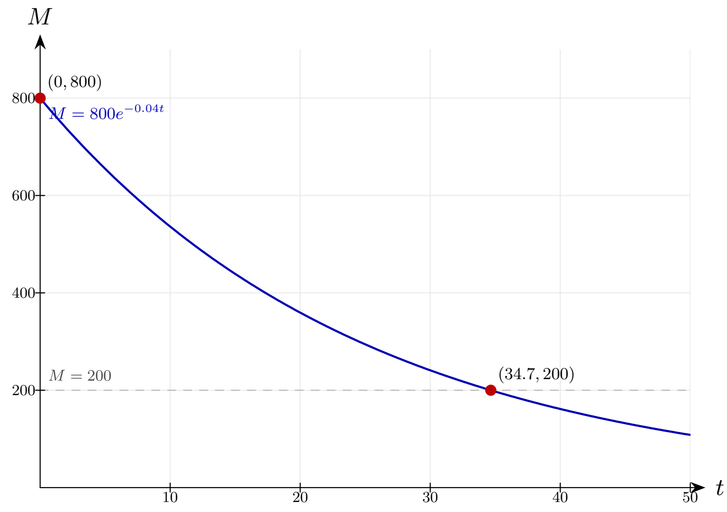
Step 3: Solve for t.

$$t = \frac{\ln 4}{0.04} \approx \frac{1.3863}{0.04} \approx 34.7 \text{ years}$$

The negatives cancel.

Final answer: $t \approx 34.7$ years

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 09 `(e15-009)`

Difficulty: Foundation | Marks: 4

A savings account is modelled by $A = 1000 \times 1.025^t$, where A is the amount (£) and t is the number of years.

Find the number of complete years required for the account to exceed £1500.

Worked solution.

Step 1: Set up the inequality.

$$1000 \times 1.025^t > 1500 \Rightarrow 1.025^t > 1.5$$

Divide both sides by 1000.

Step 2: Take log *of both sides.*

$$t \log(1.025) > \log(1.5)$$

Since $\log(1.025) > 0$, the inequality direction stays the same.

Step 3: Solve for t.

$$t > \frac{\log(1.5)}{\log(1.025)} \approx \frac{0.17609}{0.01072} \approx 16.4$$

Evaluate using a calculator.

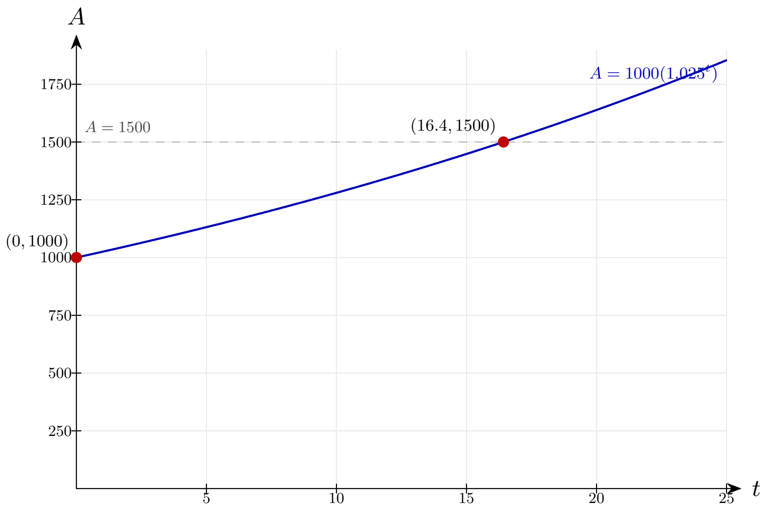
Step 4: State the number of complete years.

$$t = 17 \text{ complete years}$$

The first complete year for which $t > 16.4$ is $t = 17$.

Final answer: 17 complete years

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 10 (e15-010)

Difficulty: Foundation | *Marks: 4*

A car depreciates in value according to $V = 12000 \times 0.88^t$, where V is the value (£) and t is the time in years.

Find the number of complete years before the car is worth less than £5000.

Worked solution.

Step 1: Set up the inequality.

$$12000 \times 0.88^t < 5000 \Rightarrow 0.88^t < \frac{5000}{12000} = \frac{5}{12}$$

Divide both sides by 12000.

Step 2: Take log *of both sides. Note:* $\log(0.88) < 0$*, so the inequality reverses.*

$$t \log(0.88) < \log\left(\frac{5}{12}\right) \Rightarrow t > \frac{\log(5/12)}{\log(0.88)}$$

Dividing by a negative number reverses the inequality.

Step 3: Evaluate.

$$t > \frac{\log(5/12)}{\log(0.88)} \approx \frac{-0.3802}{-0.05552} \approx 6.85$$

Both numerator and denominator are negative, so $t > 6.85$.

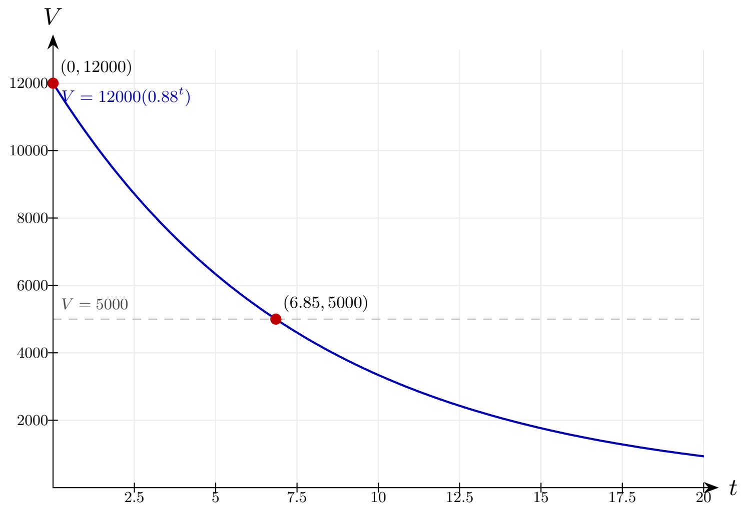
Step 4: State the answer.

$$t = 7 \text{ complete years}$$

After 7 complete years the car is first worth less than £5000.

Final answer: 7 complete years

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 11 (e15-011)

Difficulty: Foundation | Marks: 4

A drug concentration C mg/litre decays according to $C = 8e^{-0.35t}$, where t is in hours. Find the time at which the concentration first falls below 1 mg/litre. Give your answer to 3 s.f.

Worked solution.

Step 1: Set $C = 1$ and solve.

$$8e^{-0.35t} = 1 \Rightarrow e^{-0.35t} = \frac{1}{8}$$

Divide both sides by 8.

Step 2: Apply ln.

$$-0.35t = \ln\left(\frac{1}{8}\right) = -\ln 8$$

$\ln(1/8) = -\ln 8$.

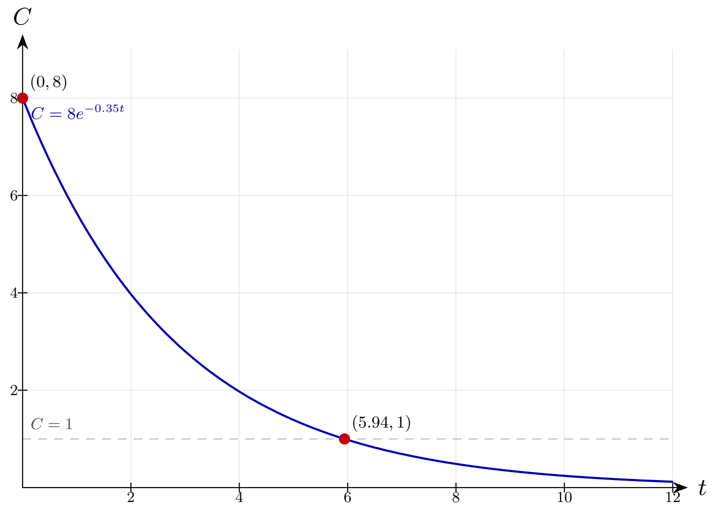
Step 3: Solve for t.

$$t = \frac{\ln 8}{0.35} \approx \frac{2.0794}{0.35} \approx 5.94 \text{ hours}$$

The negatives cancel; evaluate using a calculator.

Final answer: $t \approx 5.94$ hours

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 12 `(e15-012)`

Difficulty: Foundation | *Marks: 4*

An oven cools after being switched off. Its temperature T °C after t minutes satisfies $T = 180e^{-0.08t} + 20$.

Find the time for the oven to cool to 60°C. Give your answer to 3 s.f.

Worked solution.

Step 1: Set $T = 60$ and isolate the exponential.

$$180e^{-0.08t} + 20 = 60 \Rightarrow 180e^{-0.08t} = 40 \Rightarrow e^{-0.08t} = \frac{40}{180} = \frac{2}{9}$$

Subtract 20, then divide by 180.

Step 2: Apply ln.

$$-0.08t = \ln\left(\frac{2}{9}\right)$$

$\ln(e^{-0.08t}) = -0.08t$.

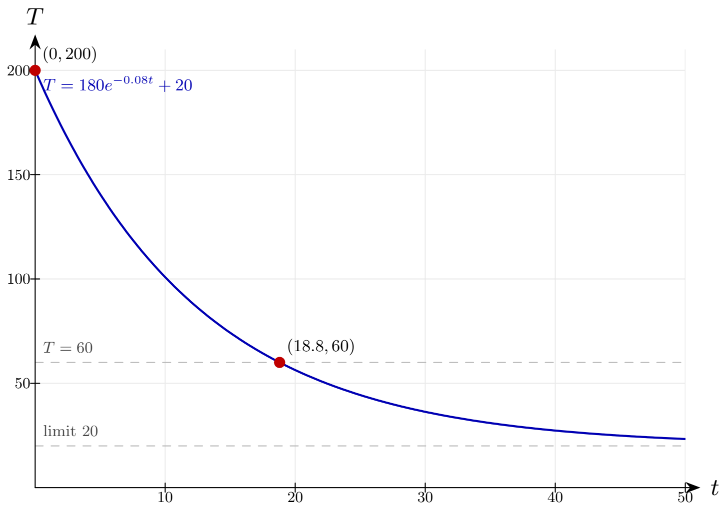
Step 3: Solve for t.

$$t = \frac{-\ln(2/9)}{0.08} = \frac{\ln(9/2)}{0.08} \approx \frac{1.5041}{0.08} \approx 18.8 \text{ min}$$

$-\ln(2/9) = \ln(9/2)$.

Final answer: $t \approx 18.8$ minutes

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 13 (e15-013)

Difficulty: Foundation | *Marks: 4*

The number of views V of a viral video after t days is modelled by $V = 500 \times 3^t$. Find the number of complete days before the video exceeds 1 000 000 views.

Worked solution.

Step 1: Set up the inequality.

$$500 \times 3^t > 1\,000\,000 \Rightarrow 3^t > 2000$$

Divide both sides by 500.

Step 2: Take log *of both sides.*

$$t \log 3 > \log 2000$$

$\log(3^t) = t \log 3$.

Step 3: Solve.

$$t > \frac{\log 2000}{\log 3} \approx \frac{3.3010}{0.4771} \approx 6.92$$

Evaluate using a calculator.

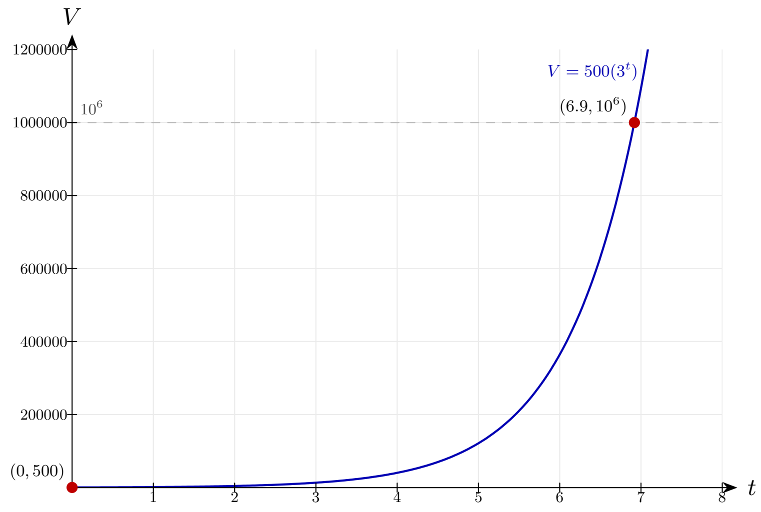
Step 4: State the number of complete days.

$$t = 7 \text{ complete days}$$

The first complete day for which $t > 6.92$ is $t = 7$.

Final answer: 7 complete days

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 14 `(e15-014)`

Difficulty: Foundation | *Marks: 4*

A population P of rabbits is modelled by $P = Ae^{kt}$. Initially there are 80 rabbits. After 5 years there are 200 rabbits.

Find the values of A and k. Give k to 3 s.f.

Worked solution.

Step 1: Substitute $t = 0$ to find A.

$$P = Ae^0 = A = 80$$

$e^0 = 1$, so $A = 80$.

Step 2: Substitute $t = 5, P = 200$.

$$80e^{5k} = 200 \Rightarrow e^{5k} = \frac{200}{80} = 2.5$$

Divide both sides by 80.

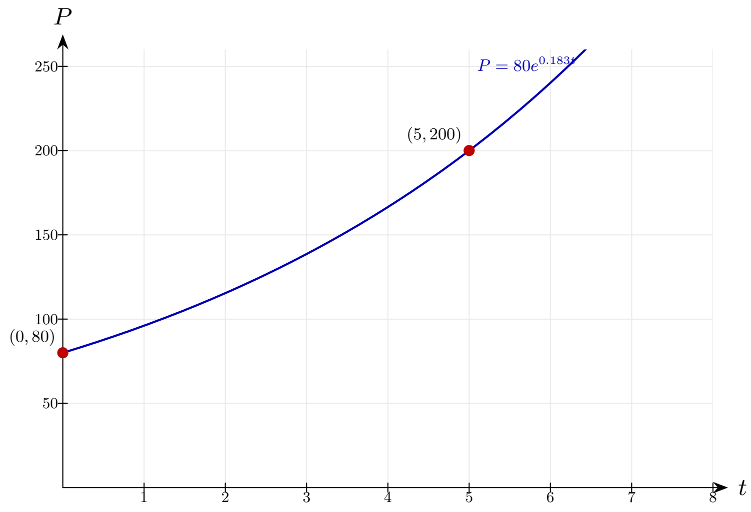
Step 3: Apply ln *and solve for k.*

$$5k = \ln(2.5) \Rightarrow k = \frac{\ln(2.5)}{5} \approx \frac{0.9163}{5} \approx 0.183$$

Divide both sides by 5.

Final answer: $A = 80,\ k \approx 0.183$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 15 `(e15-015)`

Difficulty: Foundation | *Marks: 4*

The mass M grams of a substance decays according to $M = Ae^{kt}$. The initial mass is 600 g and after 10 years the mass is 350 g.
Find A and k. Give k to 3 s.f.

Worked solution.

Step 1: Use $t = 0$ to find A.

$$M = Ae^0 = A = 600$$

The initial mass gives A directly.

Step 2: Substitute $t = 10, M = 350$.

$$600e^{10k} = 350 \Rightarrow e^{10k} = \frac{350}{600} = \frac{7}{12}$$

Divide both sides by 600.

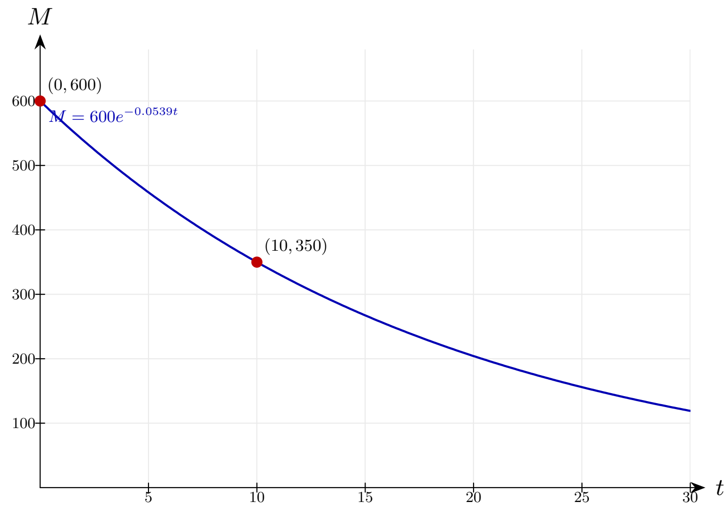
Step 3: Apply ln *and solve.*

$$10k = \ln\left(\frac{7}{12}\right) \Rightarrow k = \frac{\ln(7/12)}{10} \approx \frac{-0.5390}{10} \approx -0.0539$$

The negative value of k confirms this is decay.

Final answer: $A = 600,\ k \approx -0.0539$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 16 `(e15-016)`

Difficulty: Foundation | Marks: 5

A fungus grows according to $F = F_0 e^{gt}$. Initially it covers 3 mm^2. After 8 hours it covers 18 mm^2.
(a) Find F_0 and g. Give g to 3 s.f.
(b) Predict the area covered after 15 hours.

Worked solution.

Step 1: (a) At $t = 0$: $F_0 = 3$.

$$F_0 = 3 \text{ mm}^2$$

Initial area gives F_0 directly.

Step 2: Substitute $t = 8, F = 18$.

$$3e^{8g} = 18 \Rightarrow e^{8g} = 6$$

Divide by 3.

Step 3: Apply ln *and solve for g.*

$$g = \frac{\ln 6}{8} \approx \frac{1.7918}{8} \approx 0.224$$

Evaluate using a calculator.

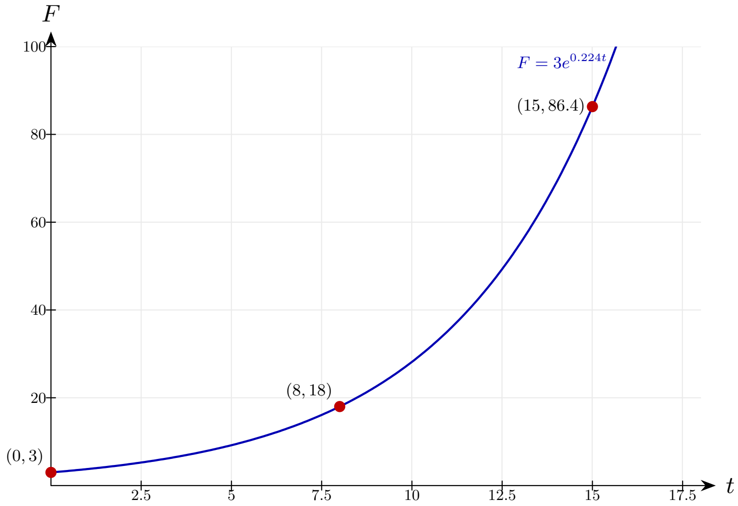
Step 4: (b) Substitute $t = 15$.

$$F = 3e^{0.224\times 15} = 3e^{3.36} \approx 3 \times 28.79 \approx 86.4 \text{ mm}^2$$

Use the values found in part (a).

Final answer: (a) $F_0 = 3,\ g \approx 0.224$ (*b*) $\approx 86.4 \text{ mm}^2$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 17 `(e15-017)`

Difficulty: Foundation | *Marks: 5*

The number of users N of a new app after t months satisfies $N = N_0 e^{kt}$. After 2 months there are 5000 users and after 6 months there are 25000 users.

Find the values of N_0 and k. Give your answers to 3 s.f.

Worked solution.

Step 1: Write two equations using the given data.

$$N_0 e^{2k} = 5000 \quad \text{and} \quad N_0 e^{6k} = 25000$$

Substitute each data point into the model.

Step 2: Divide the second equation by the first to eliminate N_0.

$$\frac{N_0 e^{6k}}{N_0 e^{2k}} = \frac{25000}{5000} \Rightarrow e^{4k} = 5$$

$e^{6k}/e^{2k} = e^{4k}$.

Step 3: Solve for k.

$$4k = \ln 5 \Rightarrow k = \frac{\ln 5}{4} \approx \frac{1.6094}{4} \approx 0.402$$

Apply ln then divide by 4.

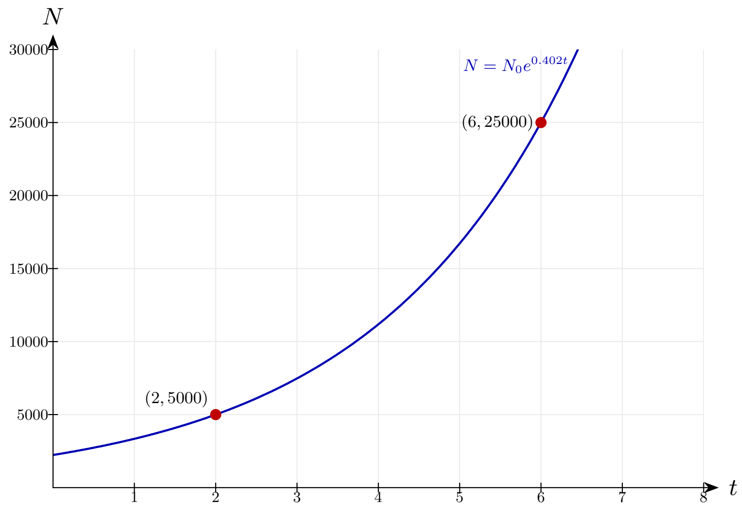
Step 4: Substitute back to find N_0.

$$N_0 = \frac{5000}{e^{2\times 0.402}} = \frac{5000}{e^{0.804}} \approx \frac{5000}{2.234} \approx 2240$$

Use the first equation with the found value of k.

Final answer: $N_0 \approx 2240$, $k \approx 0.402$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 18 `(e15-018)`

Difficulty: Foundation | *Marks: 4*

A radioactive substance has a half-life of 15 years. Its decay is modelled by $A = A_0e^{kt}$.

(a) Show that $k = -\dfrac{\ln 2}{15}$.

(b) Find the value of k to 3 s.f.

Worked solution.

Step 1: (a) The half-life means that when $t = 15$, $A = \frac{1}{2}A_0$.

$$A_0e^{15k} = \frac{1}{2}A_0 \Rightarrow e^{15k} = \frac{1}{2}$$

Divide both sides by A_0.

Step 2: Apply ln.

$$15k = \ln\left(\frac{1}{2}\right) = -\ln 2 \Rightarrow k = -\frac{\ln 2}{15}$$

$This completes the proof.$✓

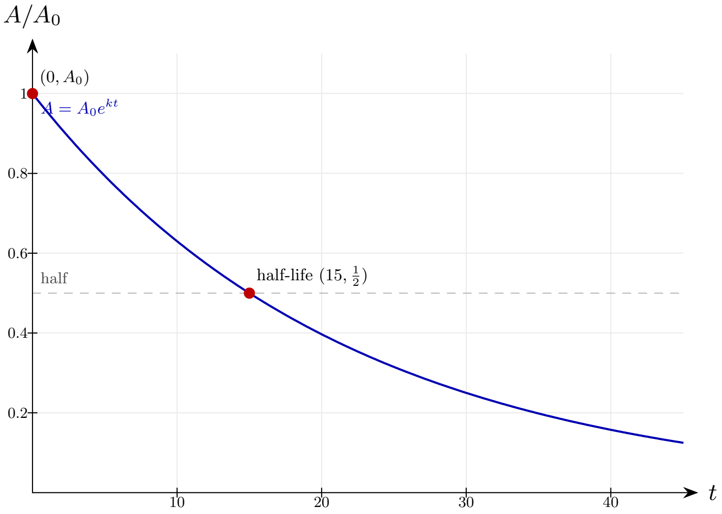
Step 3: (b) Evaluate numerically.

$$k = -\frac{\ln 2}{15} \approx -\frac{0.6931}{15} \approx -0.0462$$

Use a calculator.

Final answer: (b) $k \approx -0.0462$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 19 (e15-019)

Difficulty: Foundation | *Marks: 6*

A radioactive substance has a half-life of 20 years and an initial activity of 640 Bq. Its activity is modelled by $A = A_0e^{kt}$.

(a) Find k to 3 s.f.

(b) Find the activity after 30 years.

(c) After how many years does the activity first fall below 50 Bq? Give your answer to 3 s.f.

Worked solution.

Step 1: (a) Use the half-life condition: $A = 320$ when $t = 20$.

$$640e^{20k} = 320 \Rightarrow e^{20k} = 0.5 \Rightarrow k = \frac{\ln(0.5)}{20} = -\frac{\ln 2}{20} \approx -0.0347$$

$\ln(0.5) = -\ln 2$.

Step 2: (b) Substitute $t = 30, A_0 = 640, k \approx -0.0347$.

$$A = 640e^{-0.0347\times30} = 640e^{-1.041} \approx 640 \times 0.3536 \approx 226 \text{ Bq}$$

Evaluate using a calculator.

Step 3: (c) Set $A = 50$ and solve.

$$640e^{-0.0347t} = 50 \Rightarrow e^{-0.0347t} = \frac{50}{640} \approx 0.07813$$

Divide both sides by 640.

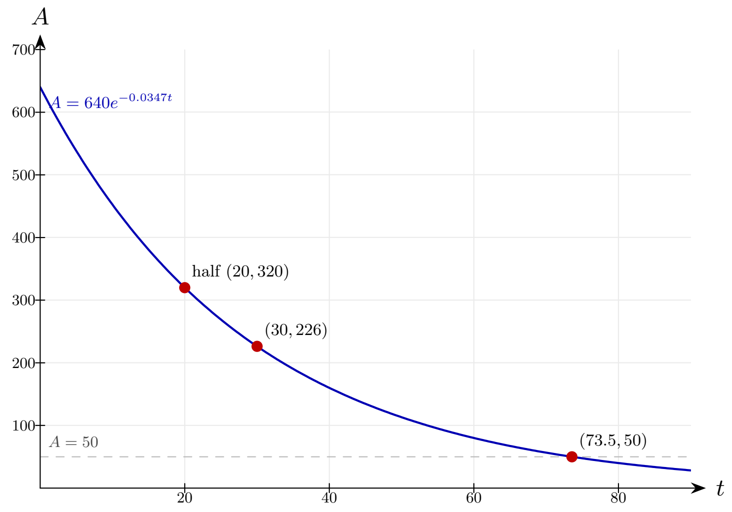
Step 4: Apply $\ln$ and solve.

$$t = \frac{\ln(50/640)}{-0.0347} = \frac{-2.5507}{-0.0347} \approx 73.5 \text{ years}$$

Both numerator and denominator are negative; $t > 0$.

Final answer: (a) $k \approx -0.0347$ (b) ≈ 226 Bq (c) ≈ 73.5 years

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 20 (e15-020)

Difficulty: Foundation | Marks: 3

A substance has a half-life of 8 years. The initial activity is 1000 Bq. After how many years is the substance reduced to a quarter of its original activity?

Worked solution.

Step 1: Recognise that a quarter of the original activity means two half-lives have passed.

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

After one half-life: $\frac{1}{2}$. After two half-lives: $\frac{1}{4}$.

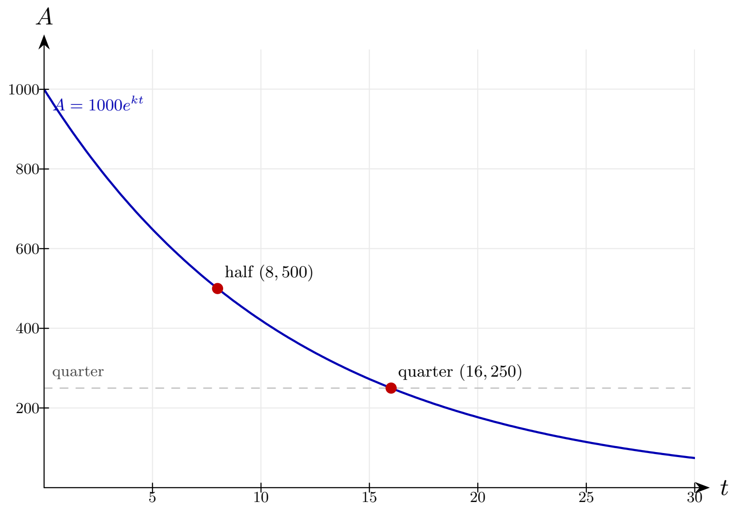
Step 2: Calculate the time for two half-lives.

$$2 \times 8 = 16 \text{ years}$$

Each half-life is 8 years.

Final answer: 16 years

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 21 `(e15-021)`

Difficulty: Foundation | Marks: 4

The temperature T ℃ of a room t minutes after the heating is turned on is modelled by $T = 22 - 16e^{-0.05t}$.

(a) State the initial temperature of the room.

(b) Explain what happens to T as $t \to \infty$.

(c) Explain why this model has an upper limit and state what it is.

Worked solution.

Step 1: (a) Substitute $t = 0$.

$$T = 22 - 16e^0 = 22 - 16 = 6\,℃$$

The room starts at 6°C.

Step 2: (b) As $t \to \infty$, $e^{-0.05t} \to 0$.

$$T \to 22 - 16(0) = 22\,℃$$

The temperature approaches 22°C but never quite reaches it.

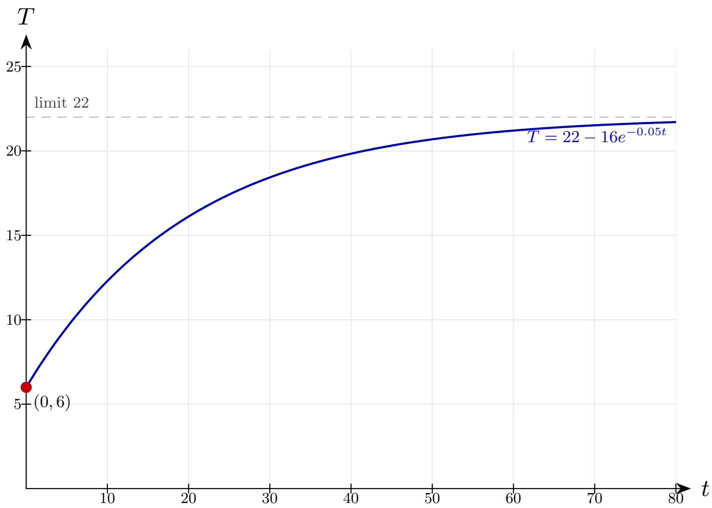
Step 3: (c) Since $e^{-0.05t} > 0$ for all finite t, the temperature never reaches 22°C.

$$T < 22\,℃ \text{ for all finite } t$$

The upper limit is 22°C. The model prevents the temperature ever exceeding this.

Final answer: (a) 6 ℃ (b) $T \to 22$ ℃ (c) Upper limit is 22 ℃

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 22 `(e15-022)`

Difficulty: Foundation | Marks: 3

The penguin population P of an island is modelled by $P = 3000e^{0.02t}$, where t is the number of years after a survey.

(a) State the initial population.

(b) Explain why this model may not be appropriate for large values of t.

Worked solution.

Step 1: (a) Substitute $t = 0$.

$$P = 3000e^{0} = 3000$$

The initial population is 3000 penguins.

Step 2: (b) Evaluate for a large t, e.g. $t = 200$.

$$P = 3000e^{0.02\times 200} = 3000e^{4} \approx 163\,794$$

After 200 years the model predicts over 163 000 penguins.

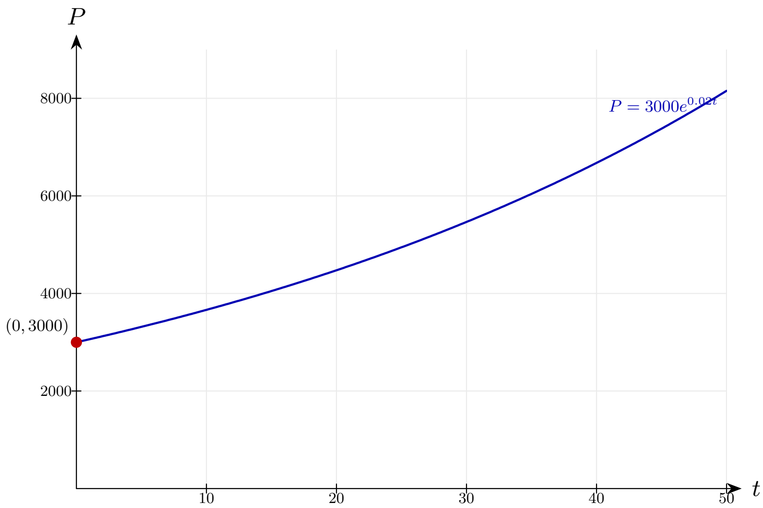
Step 3: Explain the limitation.

$$\text{The model allows } P \to \infty \text{ but island resources are finite.}$$

In reality, food supply, predators and habitat would limit growth. The model ignores these factors and is unrealistic in the long term.

> **Final answer:** (a) 3000 (b) The model allows unlimited growth, which is unrealistic — food supply and habitat would limit the population in reality.

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 23 (e15-023)

Difficulty: Foundation | *Marks: 6*

A car's value (£) t years after purchase is modelled by $V = 2000 + 11000e^{-0.2t}$.

(a) State the purchase price.

(b) Explain the significance of the value 2000 in the model.

(c) After how many complete years is the car worth less than £4000? Give your answer to 3 s.f.

Worked solution.

Step 1: (a) Substitute $t = 0$.

$$V = 2000 + 11000e^{0} = 2000 + 11000 = £13000$$

The car was purchased for £13 000.

Step 2: (b) As $t \to \infty$, $e^{-0.2t} \to 0$, so $V \to 2000$.

$$V \to £2000$$

The 2000 represents the scrap/minimum value — the car will never be worth less than £2000 according to this model.

Step 3: (c) Set $V = 4000$.

$$2000 + 11000e^{-0.2t} = 4000 \Rightarrow 11000e^{-0.2t} = 2000 \Rightarrow e^{-0.2t} = \frac{2}{11}$$

Subtract 2000 then divide by 11000.

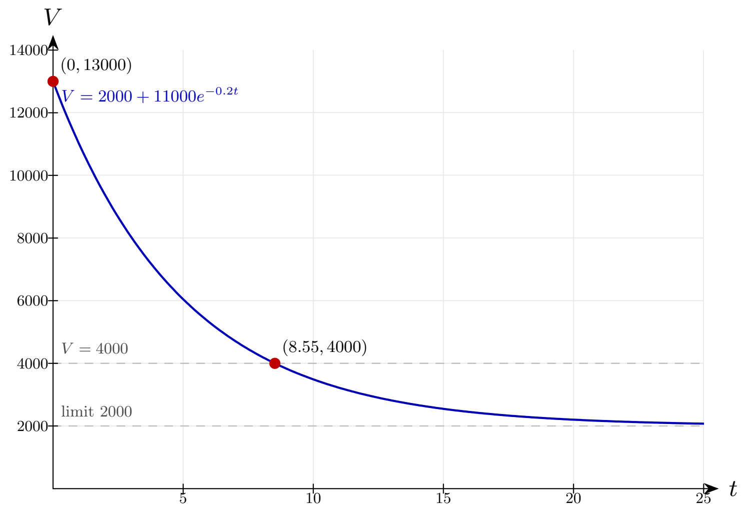
Step 4: Apply ln *and solve.*

$$-0.2t = \ln\left(\frac{2}{11}\right) \Rightarrow t = \frac{-\ln(2/11)}{0.2} = \frac{\ln(11/2)}{0.2} \approx \frac{1.7047}{0.2} \approx 8.52$$

Use a calculator; the car is worth less than £4000 after 9 complete years.

Final answer: (a) £13 000 (b) £2000 is the minimum value as $t \to \infty$ (c) After 9 complete years

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 24 (e15-024)

Difficulty: Foundation | *Marks: 5*

A forest fire spreads so that the burnt area H hectares after t hours satisfies $H = 10e^{bt}$.

(a) Interpret the value 10 in this model.

(b) After 2 hours the burnt area is 40 hectares. Find the value of b.

(c) Find the area burned after 5 hours. Give your answer to 3 s.f.

Worked solution.

Step 1: (a) Substitute $t = 0$.

$$H = 10e^0 = 10 \text{ hectares}$$

The value 10 is the initial area of the fire when first observed.

Step 2: (b) Substitute $t = 2, H = 40$.

$$10e^{2b} = 40 \Rightarrow e^{2b} = 4 \Rightarrow 2b = \ln 4 \Rightarrow b = \frac{\ln 4}{2} \approx 0.693$$

Divide by 10, apply ln, then divide by 2.

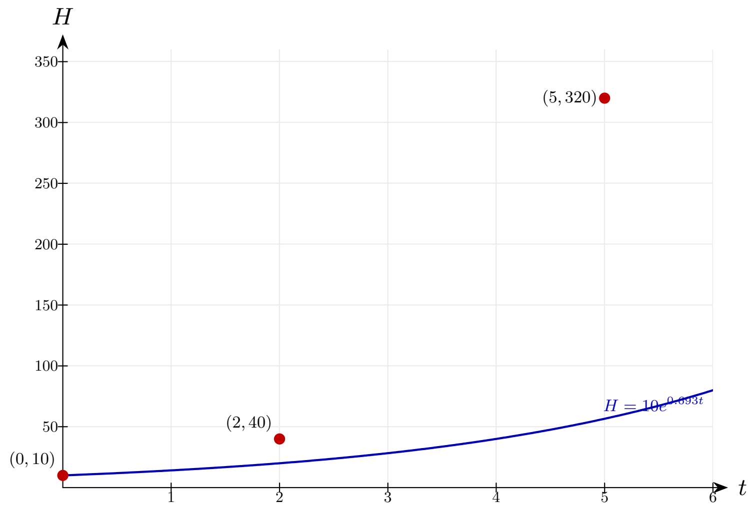
Step 3: (c) Substitute $t = 5$.

$$H = 10e^{0.693\times 5} = 10e^{3.466} \approx 10 \times 32.0 \approx 320 \text{ hectares}$$

Evaluate using a calculator.

Final answer: (a) 10 hectares is the initial area (b) $b \approx 0.693$ (c) ≈ 320 hectares

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 25 (e15-025)

Difficulty: Foundation | Marks: 2

Describe one limitation of using the model $H = 10e^{0.693t}$ to predict the area burned by a forest fire over a long period.

Worked solution.

Step 1: Consider what happens as t gets large.

$$\text{As } t \to \infty, \ H \to \infty$$

The model predicts an infinitely large burnt area.

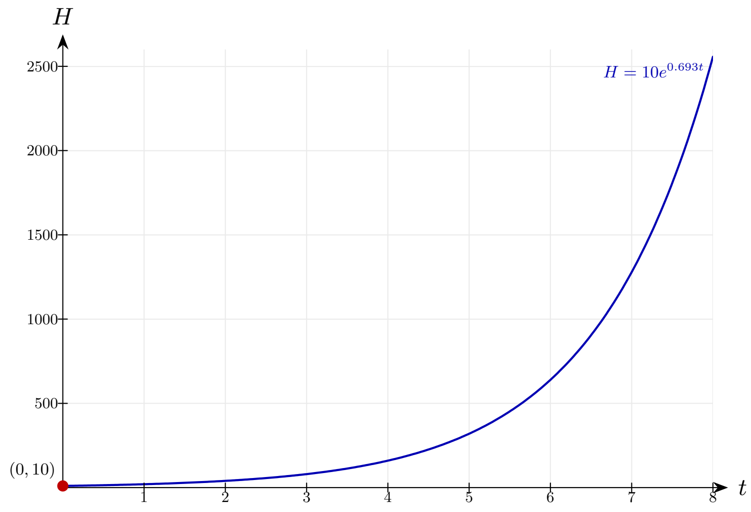
Step 2: Identify the real-world constraint.

In reality, the fire is bounded by the total area of the forest.

Once all the forest is burned, the model is no longer valid. It also ignores firefighting efforts, rainfall, and natural firebreaks.

Final answer: The model predicts unlimited growth, but the fire is constrained by the finite size of the forest. Other real-world factors (e.g. rainfall, firefighting) are also ignored.

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 26 `(e15-026)`

Difficulty: Foundation | Marks: 5

Two populations of bacteria are modelled by $B_1 = 400e^{0.03t}$ and $B_2 = 100e^{0.08t}$, where t is in hours.

(a) Which population is larger initially?

(b) Find the time at which both populations are equal. Give your answer to 3 s.f.

Worked solution.

Step 1: (a) At $t = 0$: $B_1 = 400$ and $B_2 = 100$.

$$B_1(0) = 400 > B_2(0) = 100$$

B_1 has the larger initial population.

Step 2: (b) Set $B_1 = B_2$.

$$400e^{0.03t} = 100e^{0.08t}$$

Find t when the populations are equal.

Step 3: Divide both sides by $100e^{0.03t}$.

$$4 = e^{0.05t}$$

$e^{0.08t}/e^{0.03t} = e^{0.05t}$.

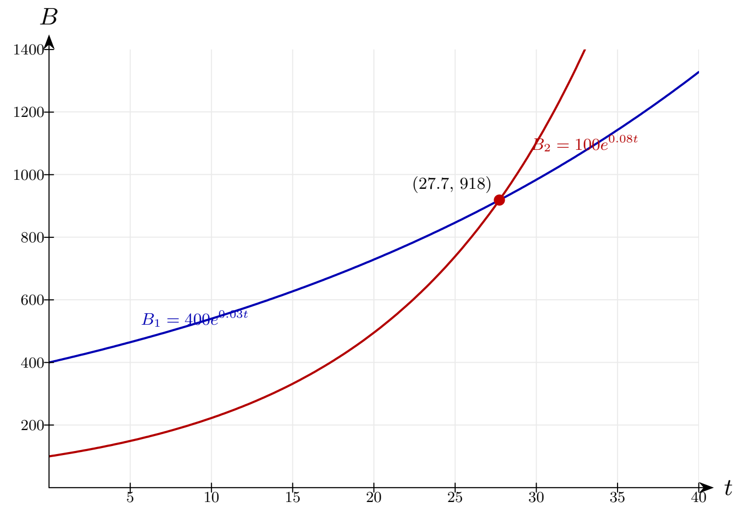
Step 4: Apply $\ln$ and solve.

$$0.05t = \ln 4 \Rightarrow t = \frac{\ln 4}{0.05} \approx \frac{1.3863}{0.05} \approx 27.7 \text{ hours}$$

Evaluate using a calculator.

Final answer: (a) B_1 is larger initially (b) $t \approx 27.7$ hours

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 27 (e15-027)

Difficulty: Foundation | Marks: 5

Two cars depreciate as follows: Car A: $V = 20000e^{-0.1t}$; Car B: $V = 15000e^{-0.06t}$, where t is in years.

(a) Which car costs more initially?

(b) After how many years are both cars worth the same amount? Give your answer to 3 s.f.

Worked solution.

Step 1: (a) At $t = 0$: Car A = £20 000, Car B = £15 000.

$$V_A(0) = £20\,000 > V_B(0) = £15\,000$$

Car A costs more initially.

Step 2: (b) Set $V_A = V_B$.

$$20000e^{-0.1t} = 15000e^{-0.06t}$$

Find when the values are equal.

Step 3: Divide both sides by $15000e^{-0.1t}$.

$$\frac{20000}{15000} = e^{(-0.06+0.1)t} \Rightarrow \frac{4}{3} = e^{0.04t}$$

$e^{-0.06t}/e^{-0.1t} = e^{0.04t}$.

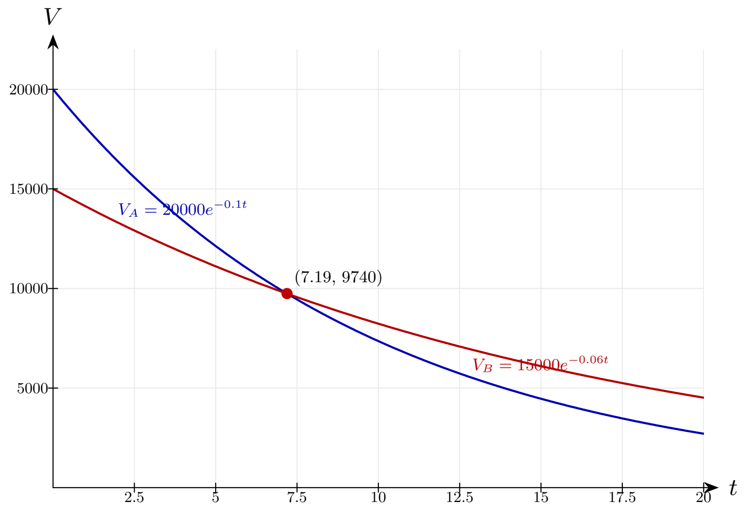
Step 4: Apply $\ln$ and solve.

$$0.04t = \ln\left(\frac{4}{3}\right) \Rightarrow t = \frac{\ln(4/3)}{0.04} \approx \frac{0.2877}{0.04} \approx 7.19 \text{ years}$$

Evaluate using a calculator.

Final answer: (a) Car A (b) $t \approx 7.19$ years

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 28 `(e15-028)`

Difficulty: Foundation | *Marks: 5*

A population of fish is modelled by $P = 600e^{0.04t}$, where t is in years.

(a) Find the rate of growth $\dfrac{dP}{dt}$ in terms of t.

(b) Evaluate the rate of growth when $t = 5$. Give your answer to 3 s.f.

(c) Show that the rate of growth is proportional to the population.

Worked solution.

Step 1: (a) Differentiate $P = 600e^{0.04t}$.

$$\frac{dP}{dt} = 0.04 \times 600e^{0.04t} = 24e^{0.04t}$$

For $y = Ae^{kt}$, the gradient is kAe^{kt}.

Step 2: (b) Substitute $t = 5$.

$$\left.\frac{dP}{dt}\right|_{t=5} = 24e^{0.2} \approx 24 \times 1.2214 \approx 29.3 \text{ fish/year}$$

Evaluate using a calculator.

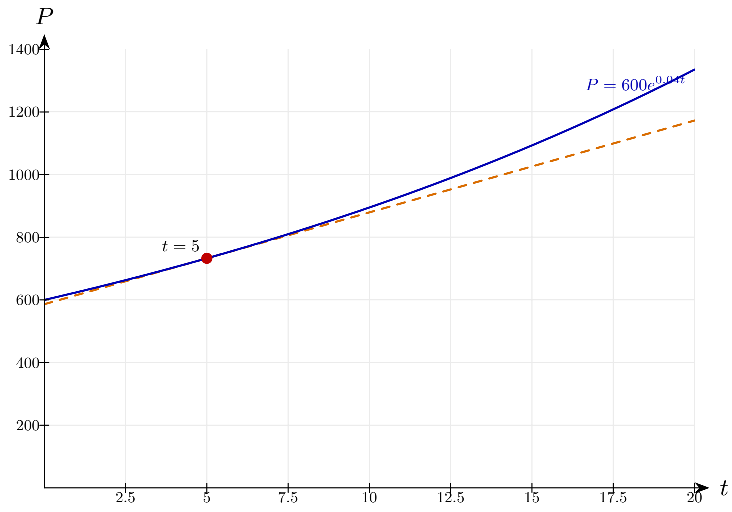
Step 3: (c) Show the ratio $\dfrac{dP/dt}{P}$ is constant.

$$\frac{dP/dt}{P} = \frac{24e^{0.04t}}{600e^{0.04t}} = \frac{24}{600} = 0.04$$

The ratio is the constant 0.04, so the rate of growth is proportional to P.✓

Final answer: (a) $\dfrac{dP}{dt} = 24e^{0.04t}$ (*b*) ≈ 29.3 fish/year

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 29 (e15-029)

Difficulty: Foundation | Marks: 4

A radioactive substance decays according to $M = 250e^{-0.05t}$ grams, where t is in years.

(a) Find the rate of decay $\frac{\mathrm{d}M}{\mathrm{d}t}$.

(b) Find the rate of decay when $t = 10$. Give your answer to 3 s.f.

Worked solution.

Step 1: (a) Differentiate.

$$\frac{\mathrm{d}M}{\mathrm{d}t} = -0.05 \times 250e^{-0.05t} = -12.5e^{-0.05t}$$

The negative sign confirms the mass is decreasing.

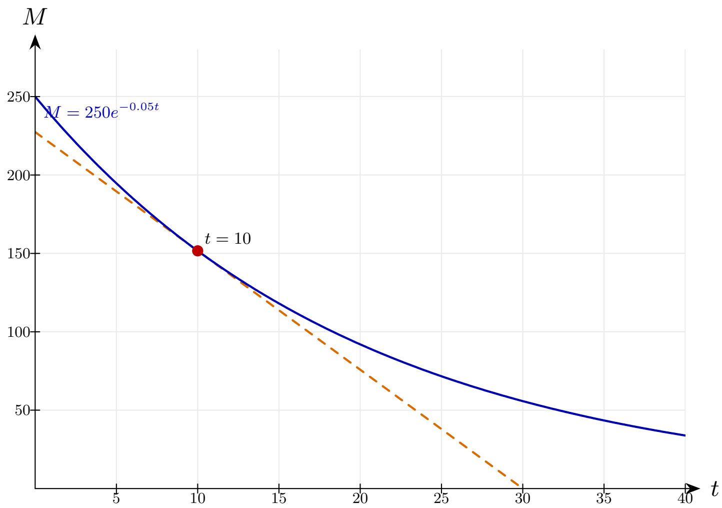
Step 2: (b) Substitute $t = 10$.

$$\left.\frac{\mathrm{d}M}{\mathrm{d}t}\right|_{t=10} = -12.5e^{-0.5} \approx -12.5 \times 0.6065 \approx -7.58 \text{ g/year}$$

The substance is losing about 7.58 g per year at $t = 10$.

Final answer: (a) $\frac{\mathrm{d}M}{\mathrm{d}t} = -12.5e^{-0.05t}$ $(b) \approx -7.58$ g/year

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 30 `(e15-030)`

Difficulty: Foundation | Marks: 6

£500 is invested in an account paying 2.5% compound interest per year. No further deposits or withdrawals are made.

(a) Write a model for the amount A (£) in the account after t years.

(b) Find the amount after 10 years. Give your answer to the nearest penny.

(c) Find the number of complete years before the account holds more than £800.

Worked solution.

Step 1: (a) Each year the amount is multiplied by 1.025.

$$A = 500 \times 1.025^t$$

$1.025 = 1 + 2.5\%$ interest rate.

Step 2: (b) Substitute $t = 10$.

$$A = 500 \times 1.025^{10} \approx 500 \times 1.2801 \approx £640.04$$

Evaluate using a calculator.

Step 3: (c) Set up the inequality.

$$500 \times 1.025^t > 800 \Rightarrow 1.025^t > 1.6$$

Divide both sides by 500.

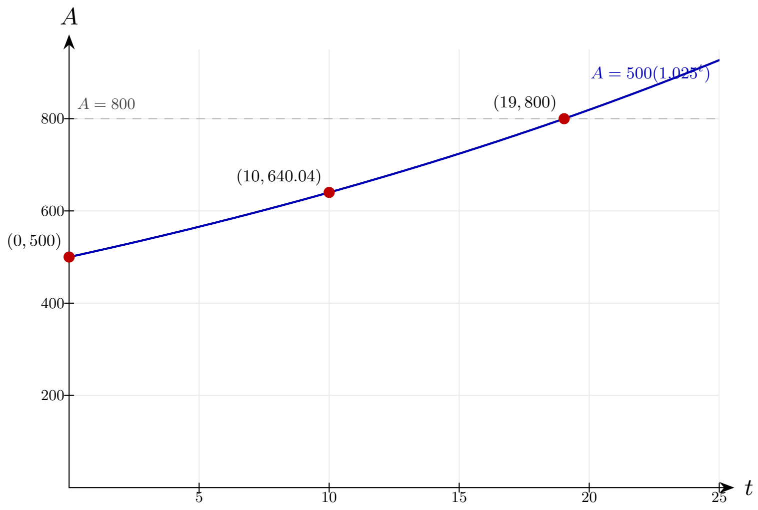
Step 4: Take logs and solve.

$$t > \frac{\log 1.6}{\log 1.025} \approx \frac{0.20412}{0.01072} \approx 19.0 \text{ years}$$

So after 20 complete years the account first exceeds £800.

Final answer: (a) $A = 500 \times 1.025^t$ (b) £640.04 (c) 20 complete years

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 31 `(e15-031)`

Difficulty: Foundation | *Marks: 7*

A patient is given a dose of medicine. The concentration C mg/litre in their bloodstream after t hours satisfies $C = Ae^{kt}$. Initially the concentration is 5 mg/litre. After 4 hours it is 2 mg/litre.

(a) Find A and k. Give k to 3 s.f.

(b) Find the concentration after 6 hours. Give your answer to 3 s.f.

(c) The medicine is effective when $C \geq 0.5$ mg/litre. Find the length of time for which the medicine is effective.

Worked solution.

Step 1: (a) At $t = 0$: $A = 5$.

$$A = 5$$

Initial condition gives A directly.

Step 2: Substitute $t = 4, C = 2$.

$$5e^{4k} = 2 \Rightarrow e^{4k} = 0.4 \Rightarrow k = \frac{\ln(0.4)}{4} \approx \frac{-0.9163}{4} \approx -0.229$$

Divide by 5, apply ln, divide by 4.

Step 3: (b) Substitute $t = 6$.

$$C = 5e^{-0.229\times 6} = 5e^{-1.375} \approx 5 \times 0.2527 \approx 1.26 \text{ mg/litre}$$

Evaluate using a calculator.

Step 4: (c) Set $C = 0.5$ and solve.

$$5e^{-0.229t} = 0.5 \Rightarrow e^{-0.229t} = 0.1 \Rightarrow -0.229t = \ln(0.1)$$

Divide by 5, apply ln.

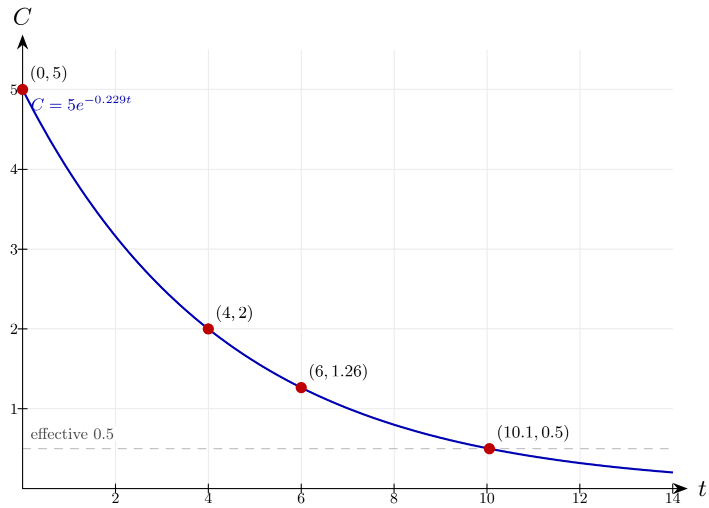
Step 5: Solve for t.

$$t = \frac{-\ln(0.1)}{0.229} = \frac{\ln 10}{0.229} \approx \frac{2.3026}{0.229} \approx 10.1 \text{ hours}$$

The medicine is effective for approximately 10.1 hours.

Final answer: (a) $A = 5,\ k \approx -0.229$ (b) ≈ 1.26 mg/litre (c) ≈ 10.1 hours

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 32 (e15-032)

Difficulty: Foundation | Marks: 6

A radioactive isotope has a half-life of 25 years. A sample initially has an activity of 480 Bq. The activity is modelled by $A = 480e^{kt}$.

(a) Show that $k = -\dfrac{\ln 2}{25}$.

(b) Find the activity after 50 years.

(c) Find the original activity if the activity after 10 years is 300 Bq. Give your answer to 3 s.f.

Worked solution.

Step 1: (a) After one half-life ($t = 25$), the activity is halved: $A = 240$.

$$480e^{25k} = 240 \Rightarrow e^{25k} = \frac{1}{2} \Rightarrow 25k = -\ln 2 \Rightarrow k = -\frac{\ln 2}{25} \checkmark$$

This proves the given expression for k.

Step 2: (b) Substitute $t = 50$ (two half-lives).

$$A = 480 \times \left(\frac{1}{2}\right)^2 = 480 \times \frac{1}{4} = 120 \text{ Bq}$$

After two half-lives, the activity is $\frac{1}{4}$ of the original.

Step 3: (c) Let the original activity be A_0. Substitute $t = 10, A = 300$.

$$A_0 e^{-\frac{\ln 2}{25} \times 10} = 300 \Rightarrow A_0 e^{-0.2773} = 300$$

Use $k = -\ln 2/25 \approx -0.02773$.

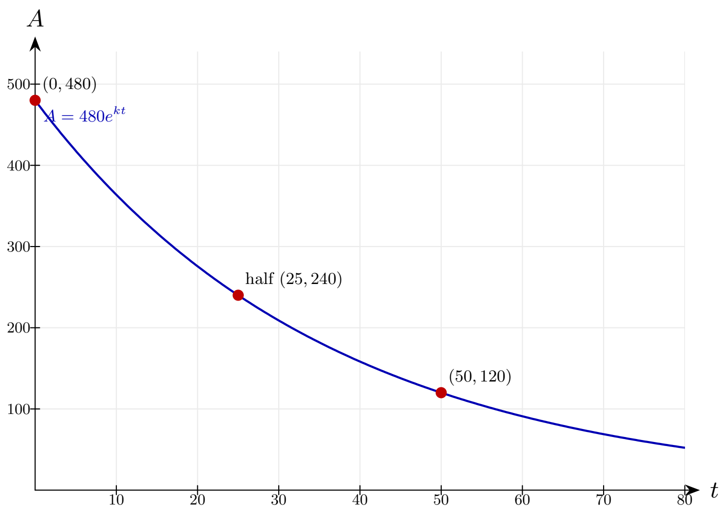
Step 4: Solve for A_0.

$$A_0 = \frac{300}{e^{-0.2773}} = 300e^{0.2773} \approx 300 \times 1.3195 \approx 396 \text{ Bq}$$

Divide by $e^{-0.2773}$ or multiply by $e^{0.2773}$.

Final answer: $(b) 120Bq \quad (c) \approx 396Bq$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 33 (e15-033)

Difficulty: Foundation | Marks: 5

A town's population P is modelled by $P = 45000e^{0.012t}$, where t is the number of years after 2020.

(a) Estimate the population in 2030.

(b) Estimate the year in which the population first exceeds 60 000.

(c) Give one reason why the model may not be reliable in the long term.

Worked solution.

Step 1: (a) In 2030, $t = 10$.

$$P = 45000e^{0.12} \approx 45000 \times 1.1275 \approx 50\,737$$

Substitute $t = 10$.

Step 2: (b) Set $P = 60000$ and solve.

$$45000e^{0.012t} = 60000 \Rightarrow e^{0.012t} = \frac{4}{3}$$

Divide both sides by 45000.

Step 3: Apply $\ln$ and solve for t.

$$t = \frac{\ln(4/3)}{0.012} \approx \frac{0.2877}{0.012} \approx 24.0$$

$t \approx 24$ years after $2020 =$ year 2044.

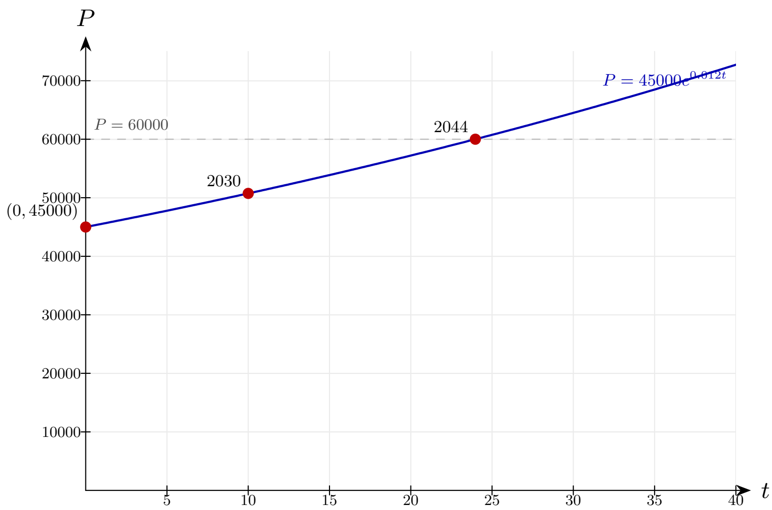
Step 4: (c) State a limitation.

Population growth depends on factors like migration, birth rate, and economic conditions.

These factors are not constant and cannot be captured by a single exponential model over a long period.

Final answer: $(a) \approx 50\,737 \quad (b) 2044 \quad (c) e.g. Assumes a constant growth rate; ignores migration and changes i$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 34 (e15-034)

Difficulty: Foundation | Marks: 7

The number of subscribers S to a YouTube channel after t weeks satisfies $S = S_0e^{kt}$. After 3 weeks there are 8000 subscribers and after 9 weeks there are 50000 subscribers.

(a) Find S_0 and k. Give k to 3 s.f.

(b) Predict the number of subscribers after 15 weeks.

(c) Describe one limitation of this model.

Worked solution.

Step 1: (a) Write two equations.

$$S_0e^{3k} = 8000 \quad \text{and} \quad S_0e^{9k} = 50000$$

Using each data point in the model.

Step 2: Divide to eliminate S_0.

$$e^{6k} = \frac{50000}{8000} = 6.25 \Rightarrow 6k = \ln(6.25) \Rightarrow k = \frac{\ln(6.25)}{6} \approx \frac{1.8326}{6} \approx 0.305$$

$e^{9k}/e^{3k} = e^{6k}$.

Step 3: Find S_0 from the first equation.

$$S_0 = \frac{8000}{e^{3\times0.305}} = \frac{8000}{e^{0.916}} \approx \frac{8000}{2.499} \approx 3202$$

Substitute k into the first equation.

Step 4: (b) Substitute $t = 15$.

$$S = 3202 \times e^{0.305\times15} = 3202e^{4.575} \approx 3202 \times 97.1 \approx 311\,000$$

Evaluate using a calculator.

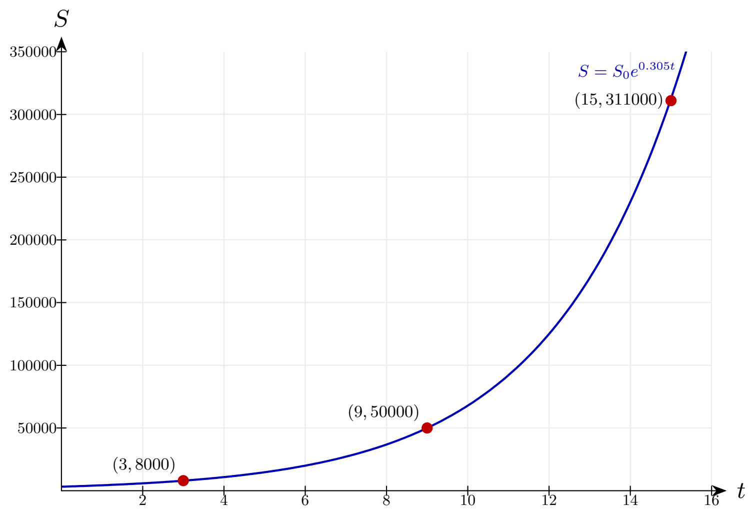
Step 5: (c) State a limitation.

The model allows $S \to \infty$ but growth will slow as the channel becomes saturated.

In reality subscriber growth slows as the potential audience is reached.

Final answer: (a) $S_0 \approx 3202,\ k \approx 0.305$ $(b) \approx 311\,000$ subscribers $(c) Unlimited growth is unrealistic; grou$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Exponential Growth and Decay 35 (e15-035)

Difficulty: Foundation | Marks: 7

A Newton's Law of Cooling model gives the temperature T ℃ of a hot drink after t minutes as $T = 65e^{-0.04t} + 20$.

(a) State the initial temperature.

(b) Find the temperature after 10 minutes. Give your answer to 3 s.f.

(c) Find the time at which the drink cools to 35°C. Give your answer to 3 s.f.

(d) Explain what the value 20 represents and state the long-term behaviour of T.

Worked solution.

Step 1: (a) Substitute $t = 0$.

$$T = 65e^{0} + 20 = 65 + 20 = 85\,℃$$

The drink starts at 85°C.

Step 2: (b) Substitute $t = 10$.

$$T = 65e^{-0.4} + 20 \approx 65 \times 0.6703 + 20 \approx 43.6 + 20 = 63.6\,^{\circ}\mathrm{C}$$

Evaluate $e^{-0.4} \approx 0.6703$.

Step 3: (c) Set $T = 35$ and isolate the exponential.

$$65e^{-0.04t} + 20 = 35 \Rightarrow 65e^{-0.04t} = 15 \Rightarrow e^{-0.04t} = \frac{15}{65} = \frac{3}{13}$$

Subtract 20 then divide by 65.

Step 4: Apply ln and solve.

$$-0.04t = \ln\left(\frac{3}{13}\right) \Rightarrow t = \frac{-\ln(3/13)}{0.04} = \frac{\ln(13/3)}{0.04} \approx \frac{1.4663}{0.04} \approx 36.7 \text{ min}$$

Evaluate using a calculator.

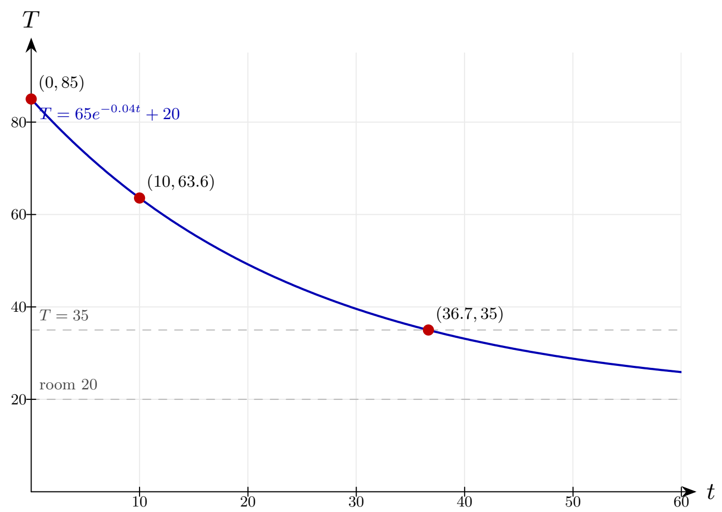
Step 5: (d) As $t \to \infty$, $e^{-0.04t} \to 0$ so $T \to 20\,^{\circ}C$.

$$T \to 20\,^{\circ}\mathrm{C}$$

The value 20 represents the room temperature — the drink cools towards the ambient temperature but never falls below it.

Final answer: (a) $85\,^{\circ}\mathrm{C}$ (b) $63.6\,^{\circ}\mathrm{C}$ (c) $t \approx 36.7$ min (d) 20°C is room temperature; $T \to 20\,^{\circ}\mathrm{C}$ as $t \to \infty$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 36 `(e15-036)`

Difficulty: Foundation | Marks: 3

A colony of bacteria doubles every 4 hours. Starting with 200, find the population after 20 hours.

Worked solution.

Step 1: Number of doublings

$$20 \div 4 = 5$$

A doubling time of 4 hours means the population is multiplied by 2 every 4 hours. Divide the total time by the doubling time to count how many doublings occur in 20 hours.

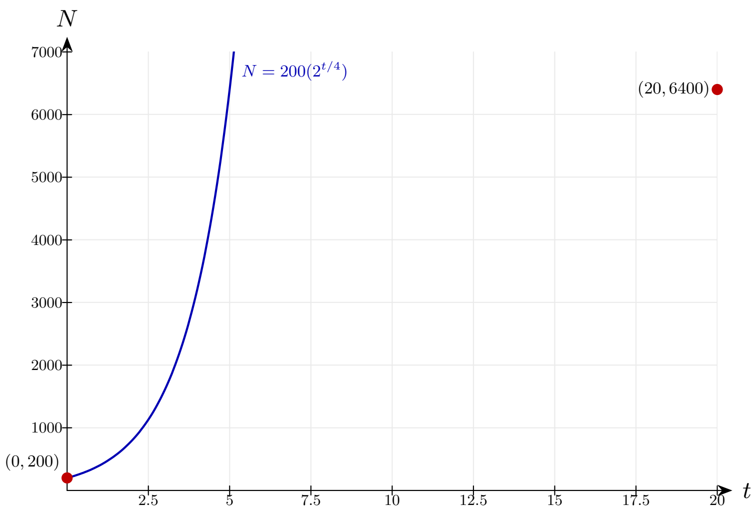
Step 2: Calculate

$$P = 200 \times 2^5 = 6400$$

Each doubling multiplies the previous population by 2, so 5 doublings multiply the starting value by $2^5 = 32$. This is the discrete form of exponential growth $P = P_0 \cdot 2^{t/T}$ with doubling time $T = 4$.

Final answer: 6400 bacteria

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 37 (e15-037)

Difficulty: Foundation | Marks: 3

A radioactive substance has half-life 5 years. If the initial mass is 80 g, find the mass after 15 years.

Worked solution.

Step 1: Number of half-lives

$$15 \div 5 = 3$$

The half-life is the time for the mass to halve. Dividing the elapsed time by the half-life counts how many halvings occur — here three.

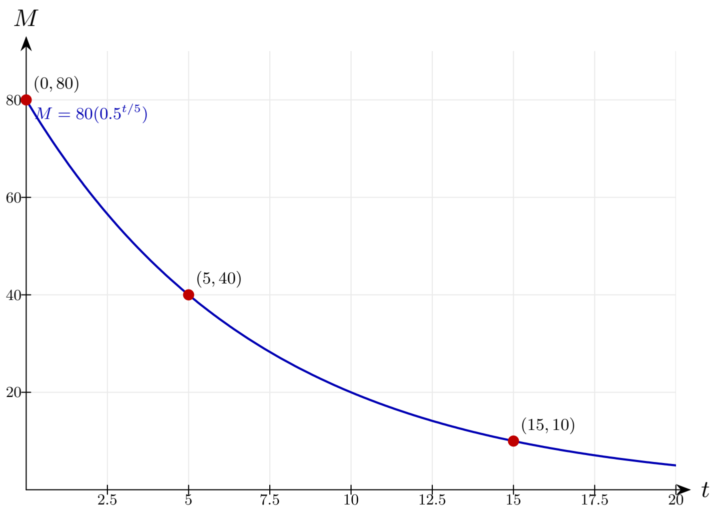
Step 2: Calculate

$$m = 80 \times \left(\frac{1}{2}\right)^3 = 10 \text{ g}$$

Each half-life multiplies the remaining mass by $\frac{1}{2}$, so after three half-lives only $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$ of the original mass remains.

Final answer: 10 g

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 38 `(e15-038)`

Difficulty: Foundation | Marks: 3

A population is modelled by $P = 1000e^{0.05t}$. Find: (a) the initial population; (b) the population after 10 years.

Worked solution.

Step 1: (a) t = 0

$$P = 1000e^{0} = 1000$$

At $t = 0$ the exponent is zero and $e^0 = 1$, so the coefficient 1000 is the initial population — the value of P at the start of the model.

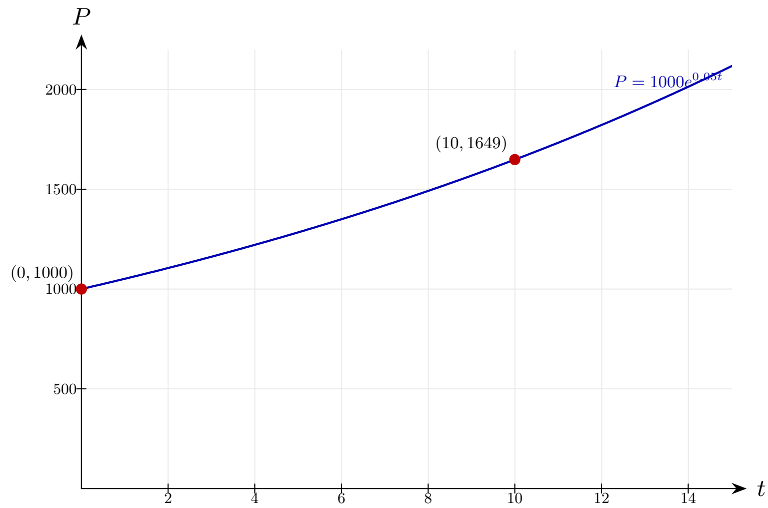
Step 2: (b) t = 10

$$P = 1000e^{0.5} = 1000 \times 1.649 = 1649$$

Substitute $t = 10$ into the model so the exponent becomes $0.05 \times 10 = 0.5$. The factor $e^{0.5} \approx 1.649$ is how much the population multiplies over 10 years, since 0.05 is the continuous growth rate per year.

Final answer: (a) 1000 (b) 1649

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 39 (e15-039)

Difficulty: Foundation | *Marks: 3*

The mass of a substance decays according to $m = 50e^{-0.1t}$. Find when the mass is 25 g.

Worked solution.

Step 1: Substitute

$$25 = 50e^{-0.1t} \implies e^{-0.1t} = 0.5$$

Setting $m = 25$ means half the original mass remains. Dividing by 50 isolates the exponential so we can apply ln cleanly in the next step.

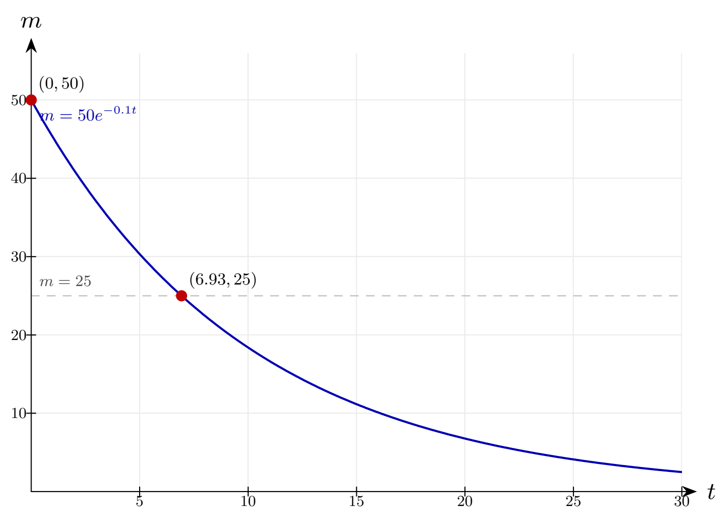
Step 2: Solve

$$t = \frac{\ln 2}{0.1} = 6.93$$

Take ln of both sides: $-0.1t = \ln(0.5) = -\ln 2$, and the negatives cancel. We use ln (not log) because the base of the exponential is e, and ln is its inverse. This is in fact the half-life of the substance.

Final answer: $t \approx 6.93$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 40 (e15-040)

Difficulty: Foundation | *Marks: 3*

A car depreciates at 15% per year. Its initial value is £20000. Find its value after 5 years.

Worked solution.

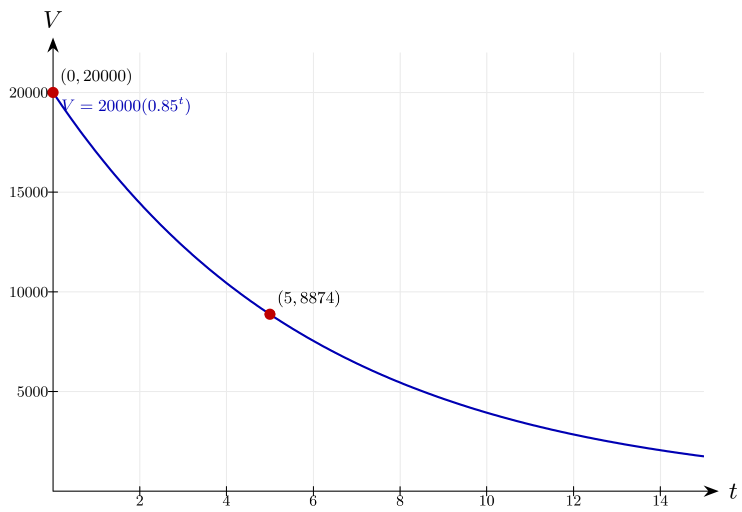
Model

$$V = 20000 \times 0.85^5 = 20000 \times 0.4437 = 8874$$

Losing 15% per year means 85% of the value carries over, so the multiplier is 0.85. Compounding for 5 years gives $0.85^5 \approx 0.4437$, meaning the car retains about 44% of its original value.

Final answer: £8874

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 41 (e15-041)

Difficulty: Foundation | *Marks: 4*

An investment grows at 3% compound interest per year. How long until it doubles?

Worked solution.

Step 1: Set up

$$1.03^n = 2$$

"Doubles" means the amount becomes $2\times$ the initial value, so after dividing by the initial amount the multiplier 1.03^n must equal 2.

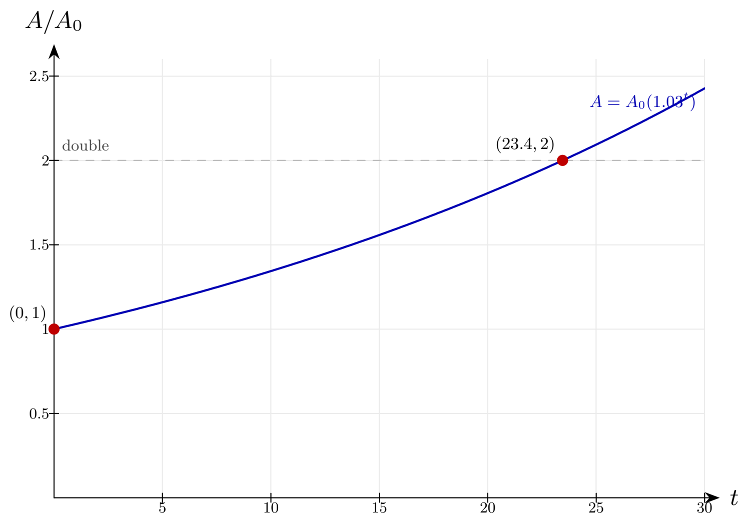
Step 2: Solve

$$n = \frac{\ln 2}{\ln 1.03} = \frac{0.693}{0.0296} = 23.4$$

Take ln of both sides: $n \ln 1.03 = \ln 2$. We use ln to bring the variable down from the exponent — any consistent log base ($\log_{10}$, ln, etc.) would work here. Since you need a whole year, round up to 24.

Final answer: Approximately 24 years

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 42 `(e15-042)`

Difficulty: Foundation | *Marks: 4*

The temperature of a cooling body is $T = 15 + 70e^{-0.03t}$. Find the initial temperature and the temperature after 30 minutes.

Worked solution.

Step 1: $t = 0$

$$T = 15 + 70 = 85°$$

At $t = 0$, $e^0 = 1$, so $T = 15 + 70$. The 70 represents the initial excess above the ambient temperature of 15°C — the body starts at 85°C.

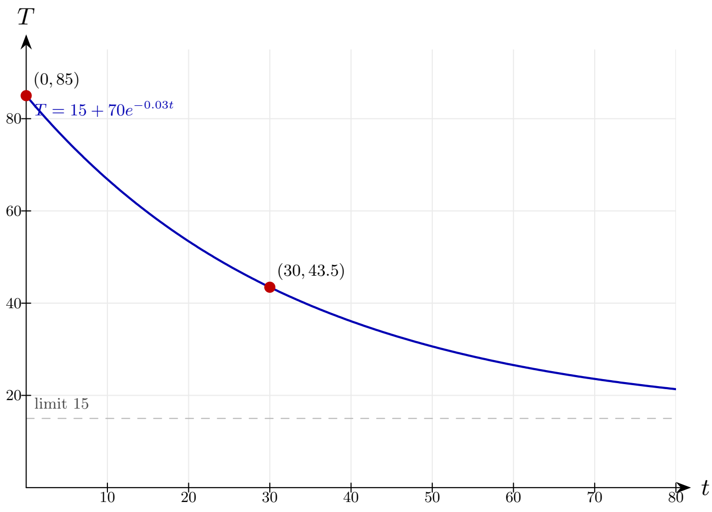
Step 2: $t = 30$

$$T = 15 + 70e^{-0.9} = 15 + 70(0.4066) = 43.5°$$

The exponent $-0.03 \times 30 = -0.9$ is negative, confirming decay: the temperature above the ambient 15°C shrinks. After 30 minutes only about 40.7% of the original excess remains, so the body sits at $15° + 28.5° \approx 43.5°$C.

Final answer: Initial: 85°C; after 30 min: 43.5°C

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 43 `(e15-043)`

Difficulty: Foundation | *Marks: 3*

A population of 5000 increases at 2% per year. Write a model and find the population after 20 years.

Worked solution.

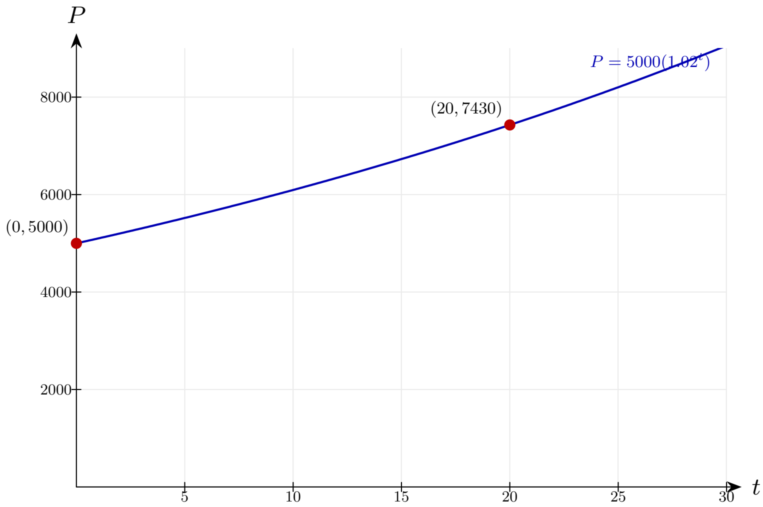
Model

$$P = 5000 \times 1.02^{20} = 5000 \times 1.486 = 7430$$

A 2% annual increase means each year the population is multiplied by 1.02 (the growth factor), so after t years $P = 5000 \times 1.02^t$. Substituting $t = 20$ gives the compounded factor $1.02^{20} \approx 1.486$, roughly a 49% increase over 20 years.

Final answer: $P = 5000 \times 1.02^t$; after 20 years: 7430

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 44 (e15-044)

Difficulty: Foundation | *Marks: 4*

A substance decays so that $m = m_0 e^{-kt}$. After 8 hours, 75% remains. Find k.

Worked solution.

Step 1: Substitute

$$0.75 = e^{-8k} \implies -8k = \ln 0.75$$

"75% remains" means $m/m_0 = 0.75$, so the exponential factor equals 0.75. Apply ln — the inverse of e — to pull the exponent down to a linear equation in k.

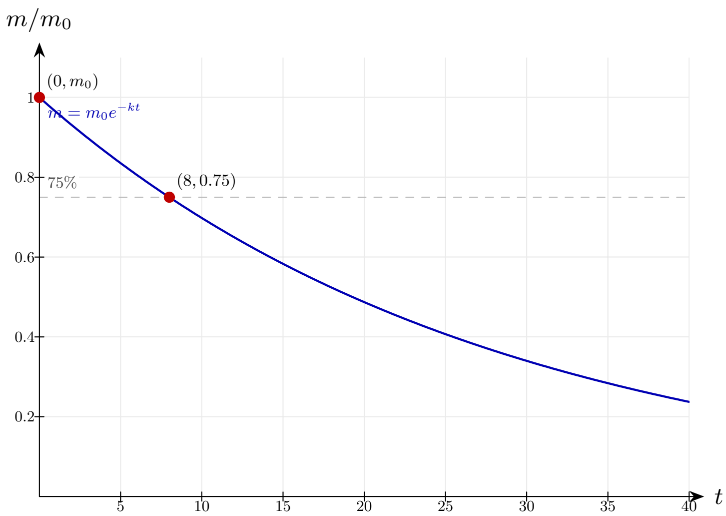
Step 2: Solve

$$k = \frac{-\ln 0.75}{8} = \frac{0.2877}{8} = 0.0360$$

Since $\ln 0.75 < 0$, the two negatives cancel and k is positive — even though it is a decay constant. The negative sign already sits in front of k in the model $m = m_0 e^{-kt}$, so a positive k corresponds to decay.

Final answer: $k \approx 0.0360$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 45 (e15-045)

Difficulty: Foundation | *Marks: 4*

The number of fish in a lake is modelled by $N = \frac{2000}{1+9e^{-0.5t}}$. Find: (a) N when t = 0; (b) N as t approaches infinity.

Worked solution.

Step 1: (a) $t = 0$

$$N = \frac{2000}{1+9} = 200$$

At $t = 0$, $e^0 = 1$, so the denominator is $1 + 9 \times 1 = 10$. This logistic model starts with 200 fish — the initial population.

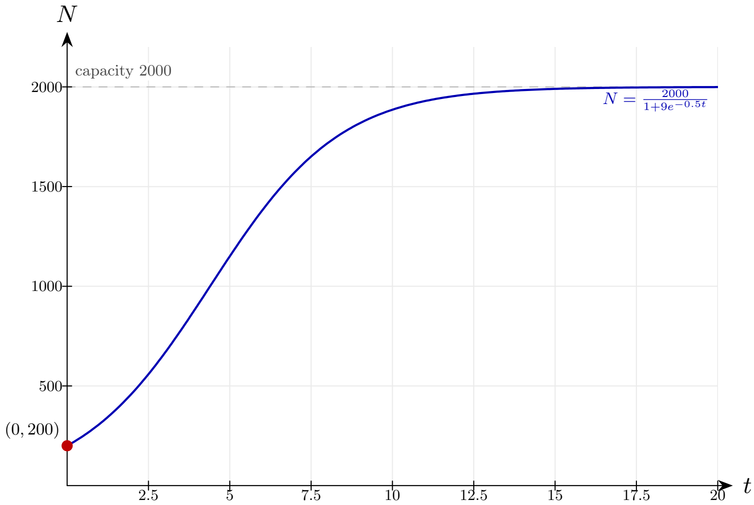
Step 2: (b) As t -> infinity

$$e^{-0.5t} \to 0 \implies N \to \frac{2000}{1} = 2000$$

The decaying exponential $e^{-0.5t}$ tends to 0 for large t, so the denominator approaches 1 and N approaches the numerator 2000. This limiting value is the carrying capacity — the maximum population the lake can support — and the logistic model prevents unlimited growth, unlike a pure exponential model.

Final answer: (a) 200 (b) 2000

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 46 `(e15-046)`

Difficulty: Foundation | Marks: 5

Carbon-14 has a half-life of 5730 years. A sample has 40% of its original C-14. How old is it?

Worked solution.

Step 1: Find k

$$k = \frac{\ln 2}{5730} = 0.000121$$

The standard half-life relation $k = \dfrac{\ln 2}{t_{1/2}}$ comes from setting $e^{-kt_{1/2}} = \frac{1}{2}$ and applying ln. A small k corresponds to a long half-life, so carbon-14 decays very slowly.

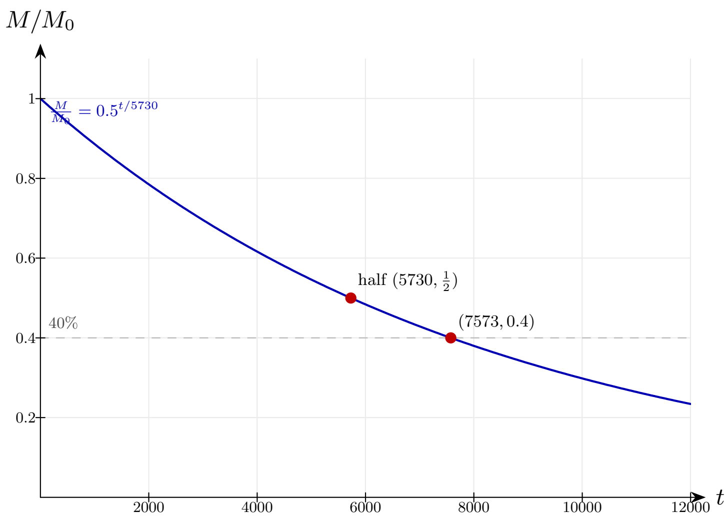
Step 2: Solve

$$0.4 = e^{-0.000121t} \implies t = \frac{-\ln 0.4}{0.000121} = 7573 \text{ years}$$

Setting the remaining fraction to 0.4 and applying ln brings the unknown t out of the exponent. The two negatives cancel because $\ln 0.4 < 0$ — and the answer is positive, consistent with time elapsed since death.

Final answer: Approximately 7573 years old

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 47 (e15-047)

Difficulty: Foundation / Marks: 4

A drug is eliminated from the body so that the amount A mg after t hours is $A = 400e^{-0.2t}$. Find when less than 50 mg remains.

Worked solution.

Step 1: Solve

$$50 = 400e^{-0.2t} \implies e^{-0.2t} = 0.125$$

Set $A = 50$ and divide by the initial dose (400) to isolate the exponential. The fraction $0.125 = \frac{1}{8}$ tells you the drug has dropped to one-eighth of the original concentration.

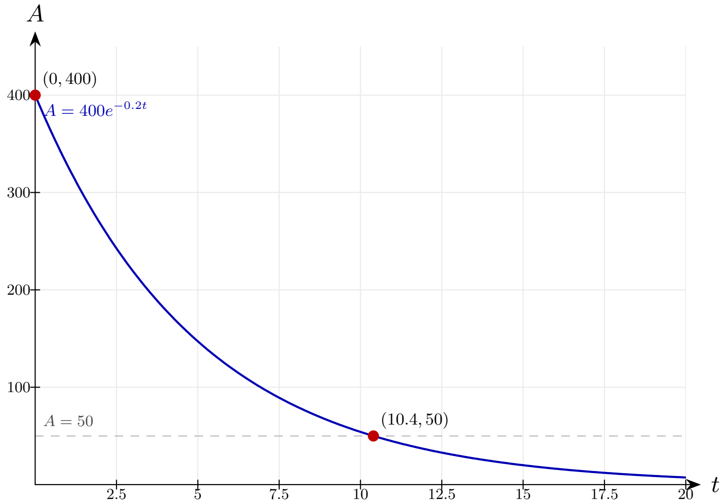
Step 2: Take ln

$$t = \frac{-\ln 0.125}{0.2} = \frac{2.079}{0.2} = 10.4 \text{ hours}$$

Apply ln to undo the e: $-0.2t = \ln(0.125) = -\ln 8$. The negatives cancel, so t is positive. Equivalently this is three half-lives (since $\frac{1}{8} = \left(\frac{1}{2}\right)^3$), each lasting $\ln 2/0.2 \approx 3.47$ hours.

Final answer: After 10.4 hours

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 48 (e15-048)

Difficulty: Foundation | *Marks: 5*

The value of a painting increases according to $V = 500e^{0.08t}$. Find: (a) the value after 10 years; (b) when it reaches £2000.

Worked solution.

Step 1: (a) t = 10

$$V = 500e^{0.8} = 500 \times 2.226 = 1113$$

Substitute $t = 10$ so the exponent is $0.08 \times 10 = 0.8$. The positive exponent confirms growth: the painting more than doubles over 10 years, reflecting the continuous growth rate of 8% per year.

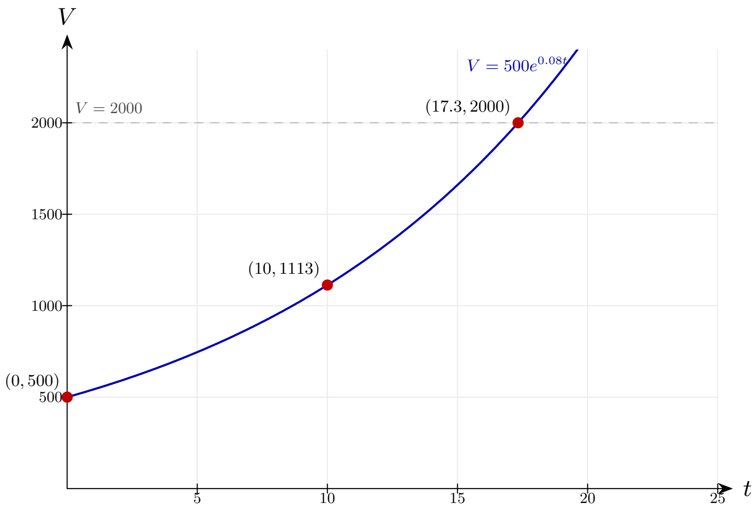
Step 2: (b) Solve

$$2000 = 500e^{0.08t} \implies e^{0.08t} = 4 \implies t = \frac{\ln 4}{0.08} = 17.3$$

Divide by 500 to isolate the exponential, then apply ln — its purpose here is to bring the exponent down so we can solve linearly for t. Since $e^{0.08t} = 4$, we have $0.08t = \ln 4$.

Final answer: (a) £1113 (b) 17.3 years

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 49 `(e15-049)`

Difficulty: Foundation | *Marks: 4*

A cup of tea cools from $90°$C in a room at $20°$C. After 5 minutes it is $70°$C. Find k in $T = 20 + 70e^{-kt}$.

Worked solution.

Step 1: Substitute $T = 70$, $t = 5$

$$70 = 20 + 70e^{-5k} \implies 50 = 70e^{-5k} \implies e^{-5k} = \frac{5}{7}$$

Subtract the room temperature (20) first — it is the ambient floor the tea cools towards, not part of the decaying excess. The remaining $70e^{-5k}$ represents the temperature above ambient at time t; dividing by 70 isolates the exponential.

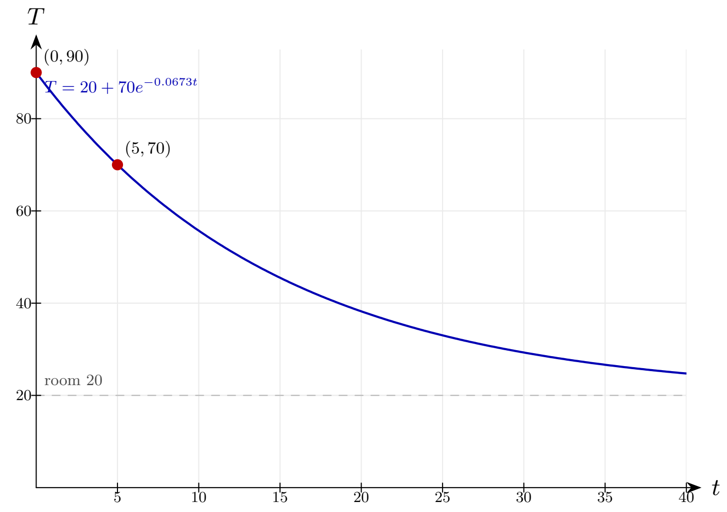
Step 2: Solve

$$k = \frac{-\ln(5/7)}{5} = \frac{0.3365}{5} = 0.0673$$

Take ln of both sides: $-5k = \ln(5/7)$. Since $5/7 < 1$, $\ln(5/7) < 0$, and the negatives cancel to give a positive k. The constant k controls how fast the tea cools — a larger k would mean a faster drop towards $20°$C.

Final answer: $k \approx 0.0673$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 50 `(e15-050)`

Difficulty: Foundation | *Marks: 5*

The spread of a rumour is modelled by $N = 500(1 - e^{-0.1t})$ where N is the number who have heard it after t hours. Find: (a) N after 5 hours; (b) when 400 people have heard it.

Worked solution.

Step 1: (a) $t = 5$

$$N = 500(1 - e^{-0.5}) = 500(1 - 0.6065) = 197$$

Substitute $t = 5$ so the exponent is $-0.1 \times 5 = -0.5$. The decaying $e^{-0.1t}$ measures the fraction yet to hear; $1 - e^{-0.1t}$ measures the fraction who have heard. The model approaches the cap of 500 as $t \to \infty$.

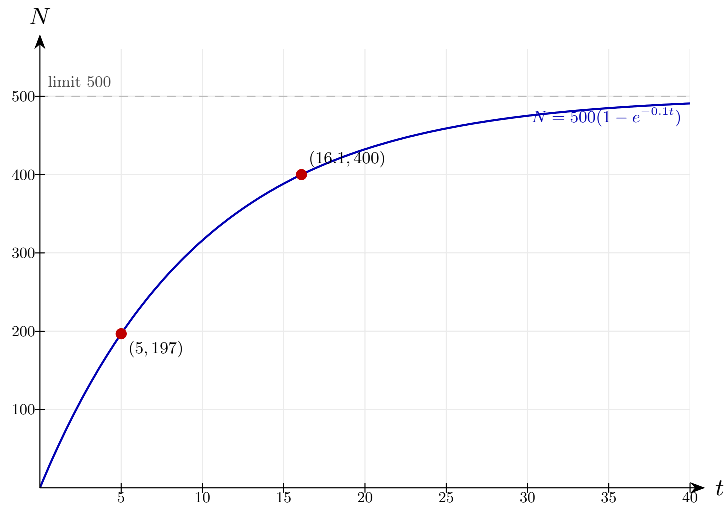
Step 2: (b) $N = 400$

$$400 = 500(1 - e^{-0.1t}) \implies e^{-0.1t} = 0.2 \implies t = \frac{\ln 5}{0.1} = 16.1$$

Divide by 500 to get $1 - e^{-0.1t} = 0.8$, then rearrange to $e^{-0.1t} = 0.2$. Apply ln: $-0.1t = \ln 0.2 = -\ln 5$. The two negatives cancel — a pattern that recurs whenever you solve a decay equation for time.

Final answer: (a) 197 people (b) 16.1 hours

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 51 `(e15-051)`

Difficulty: Foundation | *Marks: 4*

A machine loses 12% of its value each year. After how many years is it worth less than a quarter of its original value?

Worked solution.

Step 1: Model

$$0.88^n < 0.25$$

Losing 12% per year means 88% carries over, so the annual multiplier is 0.88. After n years the machine is worth $V_0 \times 0.88^n$; dividing through by V_0 leaves the condition that the multiplier falls below $\frac{1}{4}$.

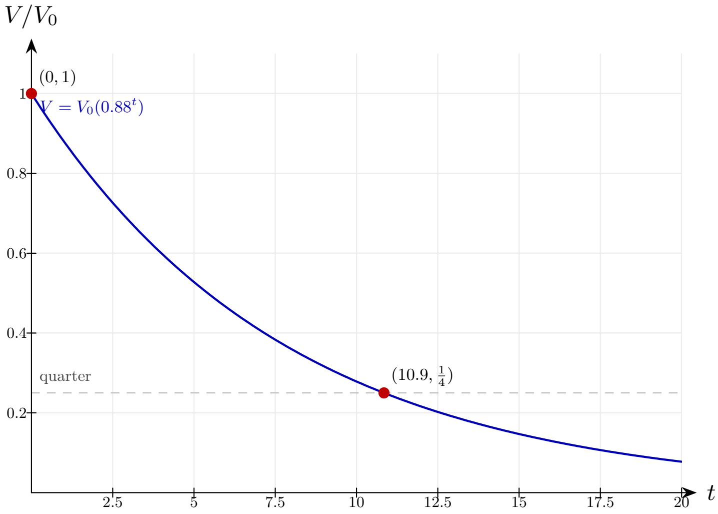
Step 2: Take logs

$$n > \frac{\ln 0.25}{\ln 0.88} = \frac{-1.386}{-0.1278} = 10.8$$

Taking ln gives $n \ln 0.88 < \ln 0.25$. Dividing by $\ln 0.88$ reverses the inequality because $\ln 0.88 < 0$ — a key pitfall in decay problems. Both ln values are negative, so the ratio is positive, and the smallest whole year is 11.

Final answer: After 11 years

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 52 (e15-052)

Difficulty: Foundation | Marks: 2

The pressure P at height h km is modelled by $P = 101e^{-0.12h}$. Find the pressure at 10 km.

Worked solution.

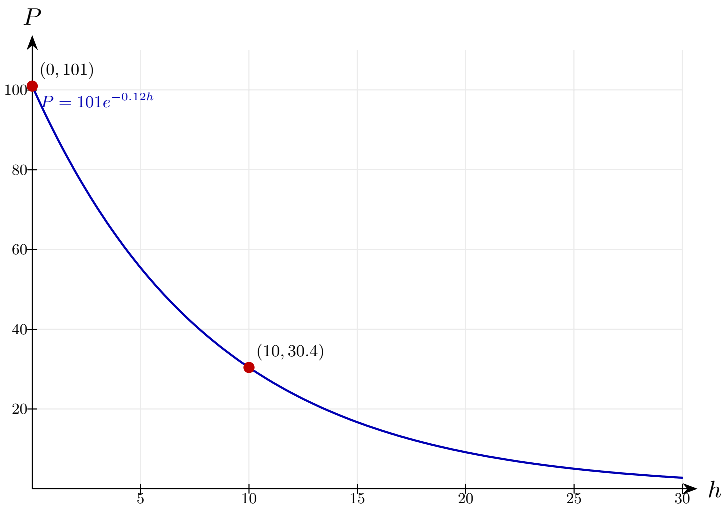
Substitute

$$P = 101e^{-1.2} = 101 \times 0.3012 = 30.4 \text{ kPa}$$

At $h = 10$ km the exponent is $-0.12 \times 10 = -1.2$. The negative exponent means pressure decays with altitude; the factor $e^{-1.2} \approx 0.30$ tells you the air at 10 km is at roughly 30% of sea-level pressure.

Final answer: 30.4 kPa

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 53 (e15-053)

Difficulty: Foundation | Marks: 3

The number of cases of a disease grows according to $N = 50e^{0.15t}$. How long until there are 500 cases?

Worked solution.

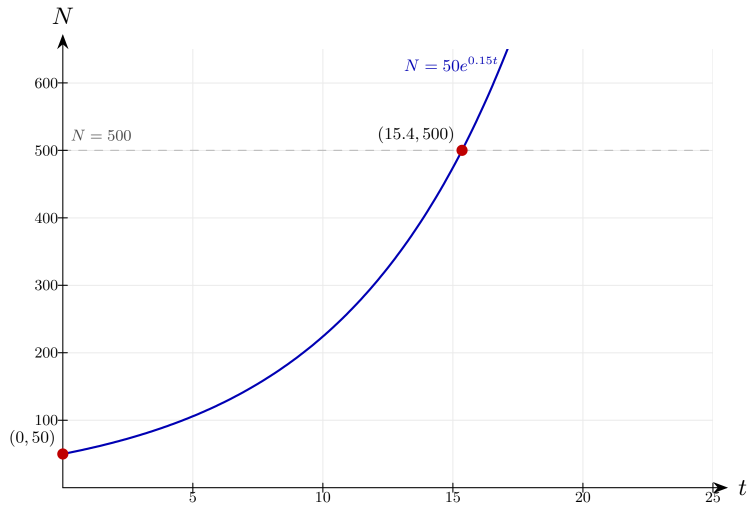
Solve

$$500 = 50e^{0.15t} \implies e^{0.15t} = 10 \implies t = \frac{\ln 10}{0.15} = 15.4 \text{ days}$$

Divide by the initial 50 cases so the exponential factor equals 10 — a tenfold increase. Apply ln to bring the time variable down from the exponent; we use ln because the base of the model is e. The positive exponent confirms growth, and $t \approx 15.4$ days gives the "tenfold time" of this outbreak.

Final answer: 15.4 days

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 54 (e15-054)

Difficulty: Foundation | Marks: 3

Show that if $y = Ae^{kt}$, then $\ln y$ is a linear function of t. State the gradient and y-intercept.

Worked solution.

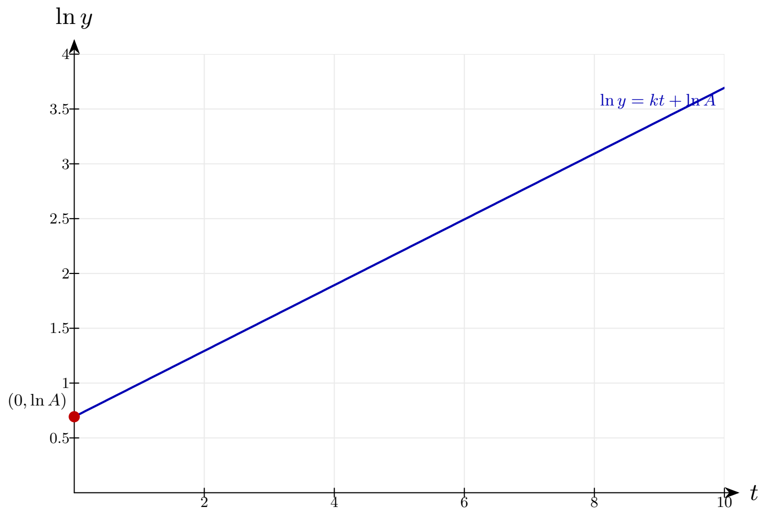
Take ln

$$\ln y = \ln A + kt$$

Apply ln to both sides and use $\ln(ab) = \ln a + \ln b$ together with $\ln(e^{kt}) = kt$. The result is linear in t with gradient k and y-intercept $\ln A$. This is why plotting $\ln y$ against t is so useful: a straight-line plot confirms exponential behaviour, and reading off the gradient gives the growth/decay constant k directly.

Final answer: Gradient $= k$, y-intercept $= \ln A$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 55 (e15-055)

Difficulty: Foundation | Marks: 3

A capacitor discharges: $V = 12e^{-t/RC}$ where $R = 1000$, $C = 0.001$. Find V after 2 seconds.

Worked solution.

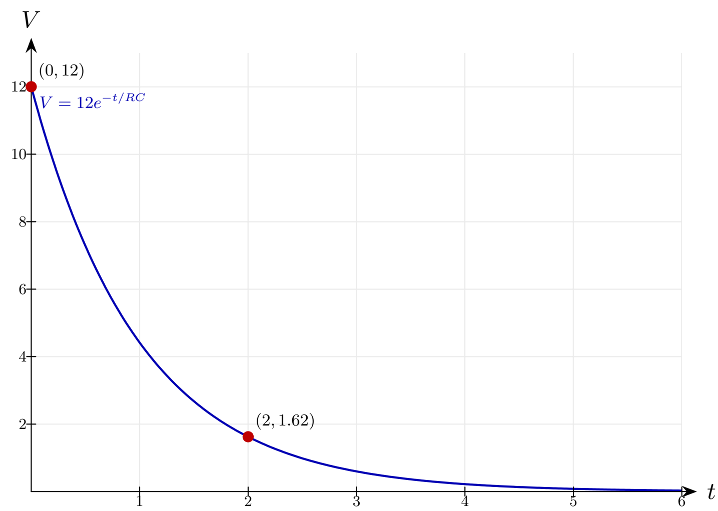
RC = 1

$$V = 12e^{-2/1} = 12e^{-2} = 12 \times 0.1353 = 1.62 \text{ V}$$

First compute the time constant: $RC = 1000 \times 0.001 = 1$ second. After $t = 2$ seconds the exponent is $-t/RC = -2$, so the capacitor retains $e^{-2} \approx 13.5\%$ of its initial voltage. The time constant RC controls how quickly the discharge happens — after one RC, about 37% remains.

Final answer: 1.62 V

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 56 (e15-056)

Difficulty: Foundation | *Marks: 4*

A town has population 8000 growing at 1.5% per year. In how many years will it exceed 10000?

Worked solution.

Step 1: Solve

$$8000 \times 1.015^n = 10000 \implies 1.015^n = 1.25$$

A 1.5% growth rate gives an annual multiplier of $1 + 0.015 = 1.015$. Dividing both sides by 8000 leaves the condition that the multiplier reaches $10000/8000 = 1.25$ — a 25% increase.

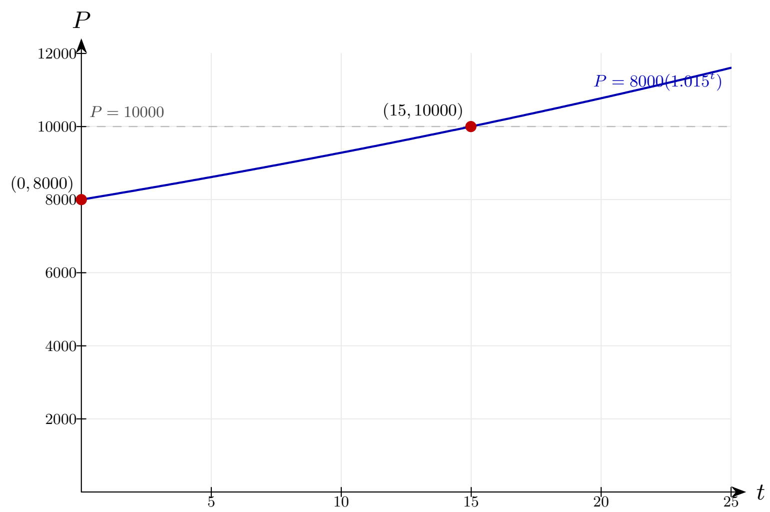
Step 2: Take logs

$$n = \frac{\ln 1.25}{\ln 1.015} = 15.0$$

Take ln of both sides to bring n down: $n \ln 1.015 = \ln 1.25$. Both ln values are positive (each base > 1), so n is positive. Population first exceeds 10 000 in the 15th year.

Final answer: 15 years

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 57 (e15-057)

Difficulty: Foundation | *Marks: 5*

Two populations: $P_1 = 3000e^{0.02t}$ and $P_2 = 5000e^{-0.01t}$. When are they equal?

Worked solution.

Step 1: Set equal

$$3000e^{0.02t} = 5000e^{-0.01t}$$

Population P_1 is growing ($+0.02$) while P_2 is decaying (-0.01), so they must cross at some future time. Setting them equal gives an equation we can solve for t.

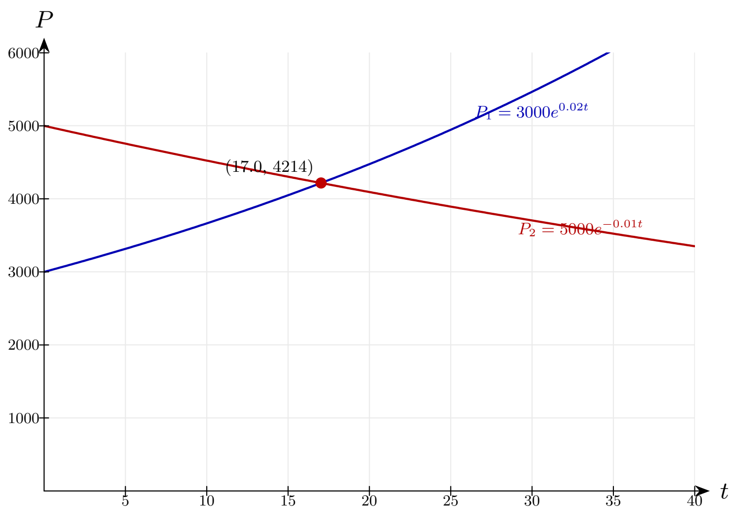
Step 2: Rearrange

$$e^{0.03t} = \frac{5}{3} \implies t = \frac{\ln(5/3)}{0.03} = 17.0$$

Divide both sides by $3000e^{-0.01t}$: the exponents combine as $0.02t - (-0.01t) = 0.03t$. Apply ln to bring t out of the exponent. The positive 0.03 reflects the net growth rate of P_1 relative to P_2.

Final answer: $t \approx 17.0 years$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 58 `(e15-058)`

Difficulty: Foundation | *Marks: 4*

A lake contains 100 kg of pollutant. Each year 20% is removed naturally. How long until less than 10 kg remains?

Worked solution.

Step 1: Model

$$100 \times 0.8^n < 10 \implies 0.8^n < 0.1$$

Removing 20% per year leaves 80%, so the annual multiplier is 0.8. Dividing by the initial 100 kg leaves the requirement that the multiplier falls below $\frac{1}{10}$.

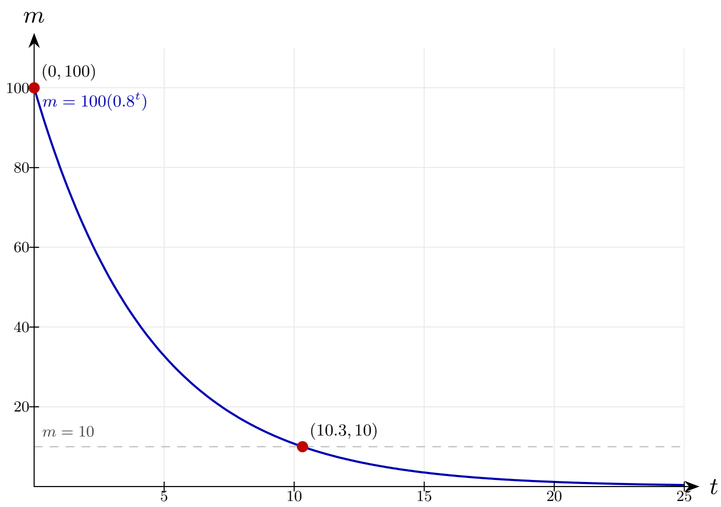
Step 2: Solve

$$n > \frac{\ln 0.1}{\ln 0.8} = \frac{-2.303}{-0.2231} = 10.3$$

Take ln of both sides: $n \ln 0.8 < \ln 0.1$. Dividing by $\ln 0.8$ reverses the inequality because $\ln 0.8 < 0$. Both ln values are negative so the ratio is positive — the smallest whole year satisfying this is 11.

Final answer: After 11 years

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 59 (e15-059)

Difficulty: Foundation | *Marks: 4*

The temperature of a freezer item placed in a room is $T = 22 - 40e^{-0.05t}$. Find when T = 10.

Worked solution.

Step 1: Substitute

$$10 = 22 - 40e^{-0.05t} \implies 40e^{-0.05t} = 12 \implies e^{-0.05t} = 0.3$$

Rearrange to isolate the exponential. The 22 is the room temperature the item is warming towards, and the 40 is the initial "deficit" below room temperature; we are looking for the moment the deficit shrinks from 40 to 12.

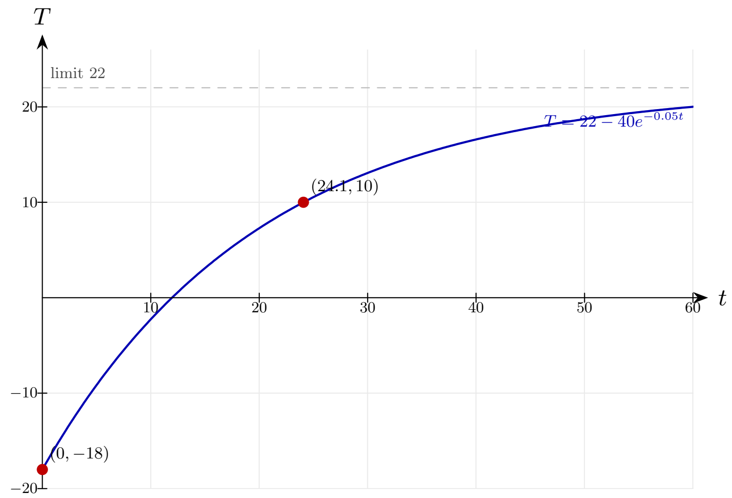
Step 2: Solve

$$t = \frac{-\ln 0.3}{0.05} = \frac{1.204}{0.05} = 24.1 \text{ min}$$

Take ln to undo the e: $-0.05t = \ln 0.3$. Since $\ln 0.3 < 0$, the negatives cancel — the standard pattern when solving decay equations for time.

Final answer: $t \approx 24.1 minutes$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 60 `(e15-060)`

Difficulty: Foundation | *Marks: 3*

A savings account pays 5% interest compounded annually. Find the annual equivalent rate if compounded continuously.

Worked solution.

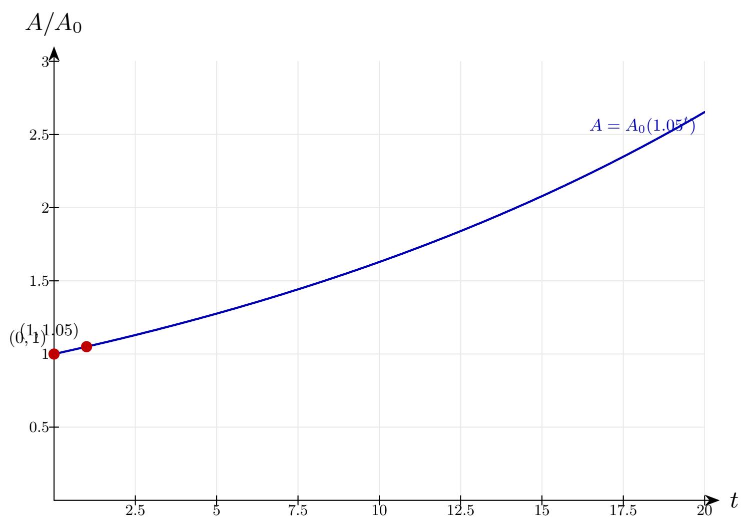
Continuous equivalent

$$e^r = 1.05 \implies r = \ln 1.05 = 0.04879$$

Annual compounding multiplies by 1.05 each year; continuous compounding multiplies by e^r per year. Setting these equal so the two schemes produce the same balance over a year gives $e^r = 1.05$, and applying ln extracts the continuous rate. The continuous rate is slightly less than 5% because continuous compounding "works harder" over the year.

Final answer: $r \approx 4.88\% continuous rate$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 61 (e15-061)

Difficulty: Foundation | Marks: 3

A rumour spreads so that after t hours, the fraction who know is $f = 1 - e^{-0.3t}$. After how many hours do 90% know?

Worked solution.

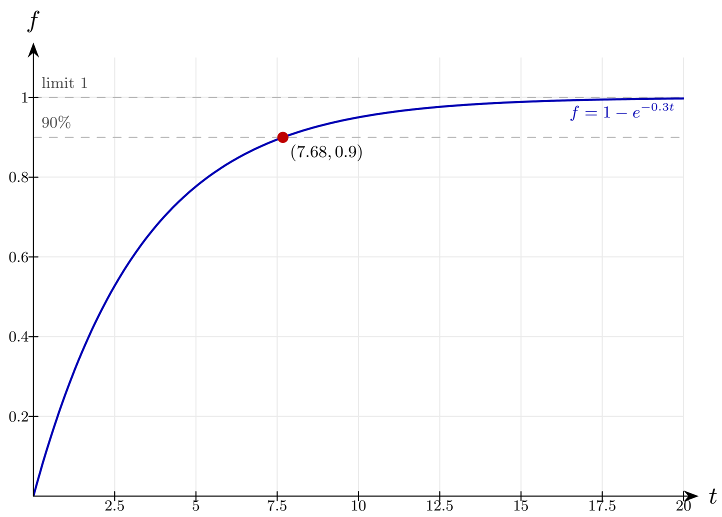
Solve

$$0.9 = 1 - e^{-0.3t} \implies e^{-0.3t} = 0.1 \implies t = \frac{\ln 10}{0.3} = 7.68$$

Set $f = 0.9$ and rearrange: $e^{-0.3t} = 1 - 0.9 = 0.1$ is the fraction who have NOT heard the rumour. Apply ln: $-0.3t = \ln 0.1 = -\ln 10$, and the negatives cancel. Notice how the model approaches but never reaches 100% — the exponential decays towards 0, so $f \to 1$ asymptotically.

Final answer: $t \approx 7.68 hours$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 62 (e15-062)

Difficulty: Foundation | Marks: 3

Light intensity decreases with depth: $I = I_0 e^{-0.4d}$ where d is in metres. At what depth is the intensity 1% of the surface value?

Worked solution.

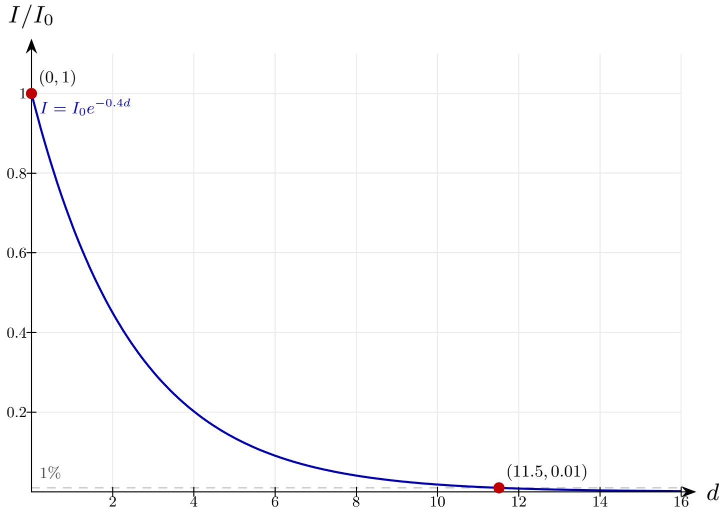
Solve

$$0.01 = e^{-0.4d} \implies d = \frac{-\ln 0.01}{0.4} = \frac{4.605}{0.4} = 11.5 \text{ m}$$

Setting $I/I_0 = 0.01$ (1% of surface intensity), apply ln to bring the depth d down from the exponent: $-0.4d = \ln 0.01 = -\ln 100$. The two negatives cancel and $d > 0$ as expected — light intensity drops with depth, which is exactly what the negative exponent $-0.4d$ encodes.

Final answer: $d \approx 11.5 metres$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 63 (e15-063)

Difficulty: Foundation | Marks: 3

The rate of decay of a substance is proportional to the amount present. Write the differential equation and its solution.

Worked solution.

Step 1: Differential equation

$$\frac{dm}{dt} = -km$$

"Rate of decay proportional to the amount present" translates directly into $\frac{dm}{dt} \propto m$. The negative sign captures that the amount is decreasing, and $k > 0$ is the decay constant.

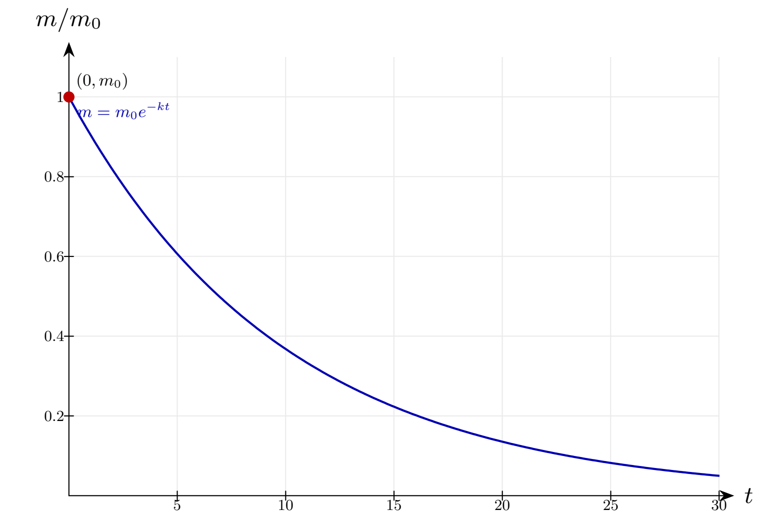
Step 2: Solution

$$m = m_0e^{-kt}$$

Separating variables and integrating gives $\ln m = -kt + C$; exponentiating recovers $m = m_0e^{-kt}$, where m_0 is the initial mass at $t = 0$. The natural exponential is the unique solution because differentiating e^{-kt} gives back $-ke^{-kt}$ — proportional to itself.

Final answer: $\frac{dm}{dt} = -km$; solution $m = m_0e^{-kt}$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 64 `(e15-064)`

Difficulty: Foundation | *Marks: 2*

House prices increase by 6% per year. A house costs £200000 now. What will it cost in 8 years?

Worked solution.

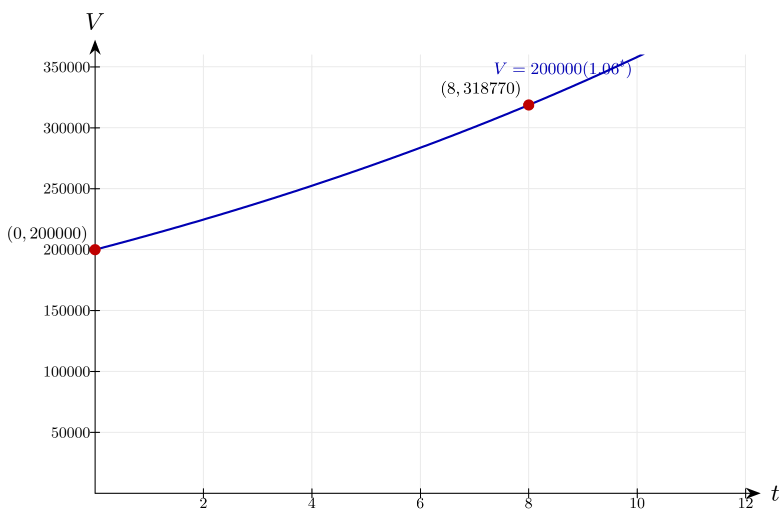
Calculate

$$V = 200000 \times 1.06^8 \approx 200000 \times 1.59385 \approx £318\,770$$

Each year multiplies the price by 1.06 (the growth factor for +6%). After 8 years that factor compounds to $1.06^8 \approx 1.59385$, so the price is roughly 59% above today's value.

Final answer: £318,770 (to the nearest pound)

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 65 (e15-065)

Difficulty: Foundation | *Marks: 3*

The charge on a capacitor builds up: $Q = Q_0(1 - e^{-t/RC})$. If $Q_0 = 100$, $RC = 5$, find Q after 10 seconds.

Worked solution.

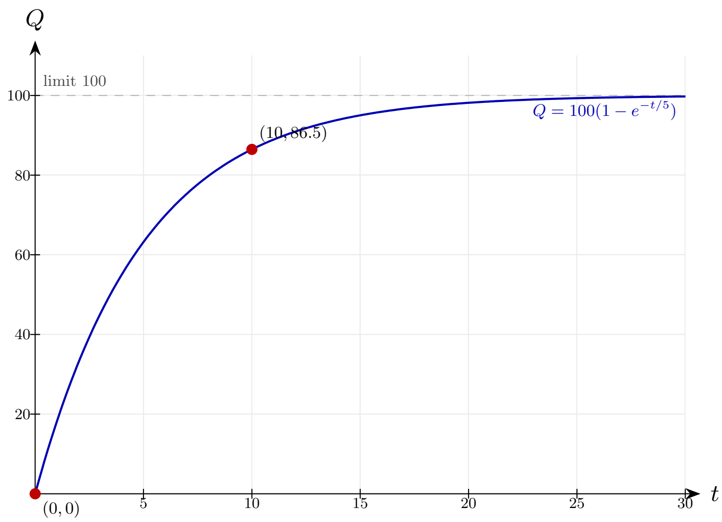
Substitute

$$Q = 100(1 - e^{-2}) = 100(1 - 0.1353) = 86.5\mu\text{C}$$

After $t = 10$ seconds the exponent is $-t/RC = -10/5 = -2$. The factor $1 - e^{-t/RC}$ measures the fraction of the maximum charge Q_0 that has built up: at $t = 2RC$ it is about 86.5%. Eventually $Q \to Q_0 = 100$ as $t \to \infty$.

Final answer: $Q \approx 86.5$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 66 (e15-066)

Difficulty: Foundation | *Marks: 4*

Data suggests $y = ab^x$. When $x = 0, y = 3$ and when $x = 4, y = 48$. Find a and b.

Worked solution.

Step 1: $x = 0$ gives a

$$a = 3$$

At $x = 0$, $b^0 = 1$ regardless of b, so $y = a$. The coefficient a is therefore the value of y at $x = 0$ — the "initial value" of the model.

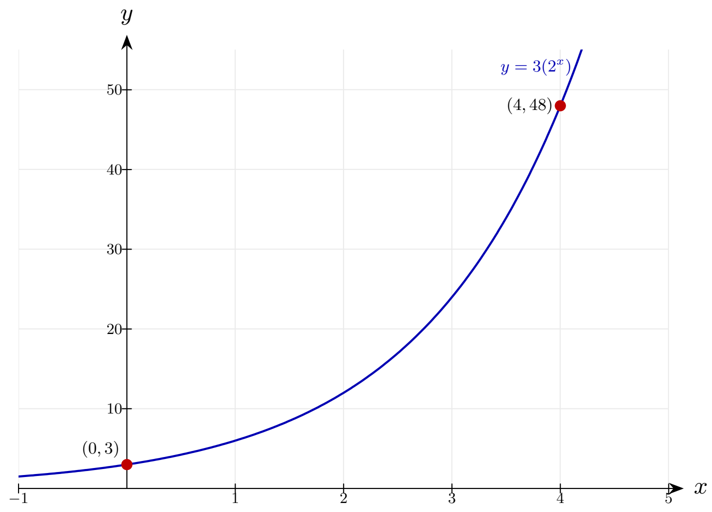
Step 2: $x = 4$

$$3b^4 = 48 \implies b^4 = 16 \implies b = 2$$

Substituting the second data point gives a single equation in b. Dividing by 3 isolates b^4, and the fourth root of 16 is 2. The base b is the per-unit growth factor: $b = 2$ means y doubles each time x increases by 1.

Final answer: $a = 3,\ b = 2$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 67 (e15-067)

Difficulty: Foundation | Marks: 4

A ball bounces to 80% of its previous height. Dropped from 2 m, after how many bounces is it below 0.5 m?

Worked solution.

Step 1: Model

$$2 \times 0.8^n < 0.5 \implies 0.8^n < 0.25$$

Each bounce multiplies the height by 0.8, so after n bounces the height is 2×0.8^n. Dividing through by 2 (the drop height) leaves the requirement that the multiplier falls below $\frac{1}{4}$.

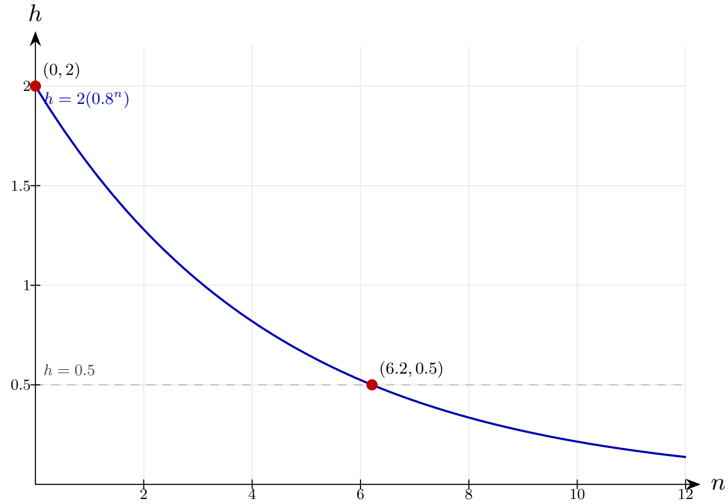
Step 2: Solve

$$n > \frac{\ln 0.25}{\ln 0.8} = \frac{-1.386}{-0.2231} = 6.21$$

Apply ln to both sides: $n \ln 0.8 < \ln 0.25$. The inequality flips when dividing by $\ln 0.8 < 0$. Both ln values are negative so the answer is positive, and we round up to the next whole bounce: 7.

Final answer: After 7 bounces

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 68 (e15-068)

Difficulty: Foundation | Marks: 3

Show that a quantity with constant percentage growth rate $r\%$ per unit time can be written as $Q = Q_0 e^{kt}$ where $k = \ln(1 + r/100)$.

Worked solution.

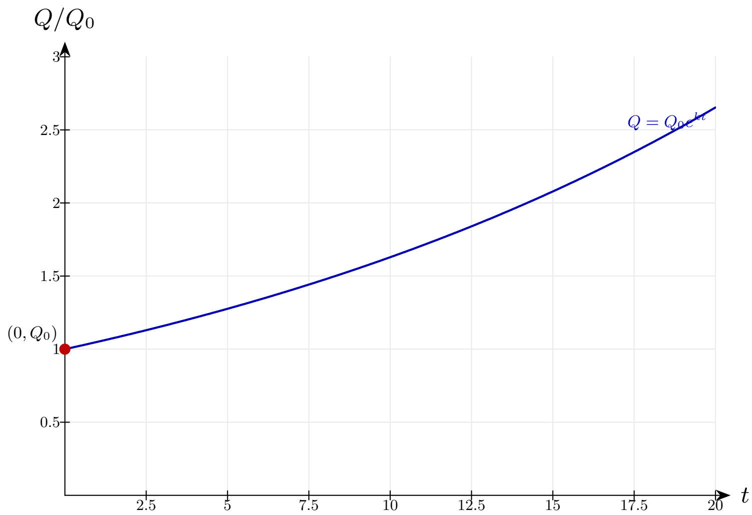
Start from discrete

$$Q = Q_0(1 + r/100)^t = Q_0 e^{t \ln(1+r/100)} = Q_0 e^{kt}$$

Use the identity $a^t = e^{t \ln a}$ to rewrite the discrete compounding factor as a natural exponential. Matching exponents with $Q_0 e^{kt}$ gives $k = \ln(1 + r/100)$. This is why "percentage growth rate" r and "continuous rate" k are not identical — k is slightly smaller than $r/100$ for positive growth.

Final answer: Shown: $k = \ln(1 + r/100)$

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 69 (e15-069)

Difficulty: Foundation | *Marks: 5*

Bacteria grow from 100 to 300 in 6 hours. Assuming exponential growth $N = 100e^{kt}$, find: (a) k; (b) the population after 10 hours.

Worked solution.

Step 1: (a) Find k

$$300 = 100e^{6k} \implies k = \frac{\ln 3}{6} = 0.183$$

Divide by 100 (the initial count) so the exponential equals 3 — the population has tripled. Apply ln to bring $6k$ down, then divide by 6. The positive k confirms growth.

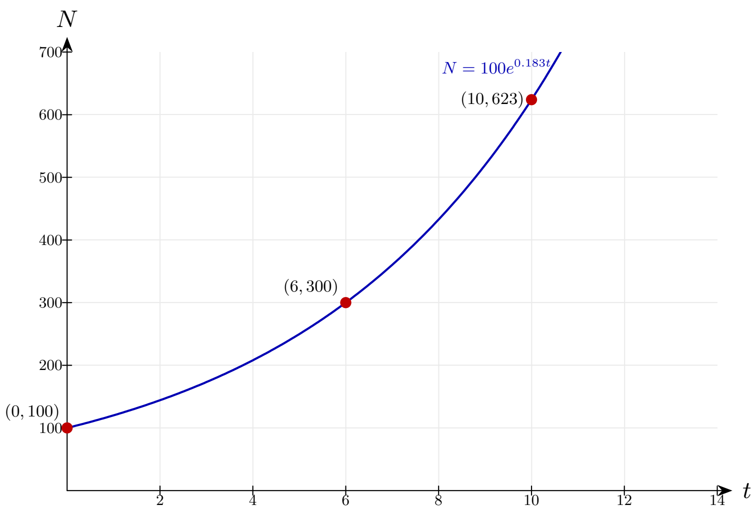
Step 2: (b) $t = 10$

$$N = 100e^{1.83} = 100 \times 6.23 = 623$$

Substitute $t = 10$ so the exponent becomes $10k \approx 1.83$. The factor $e^{1.83} \approx 6.23$ is how much the colony multiplies in 10 hours under the fitted growth rate.

Final answer: (a) $k \approx 0.183$ (b) 623 bacteria

Diagram.



The model plotted against time, with the key value(s) marked.

Modelling Growth/Decay 70 (e15-070)

Difficulty: Foundation | *Marks: 7*

A radioactive element has decay constant $k = 0.005$ per year. Find: (a) the half-life; (b) the time for 90% to decay; (c) the percentage remaining after 100 years.

Worked solution.

Step 1: (a) Half-life

$$t_{1/2} = \frac{\ln 2}{0.005} = 138.6 \text{ years}$$

The half-life satisfies $e^{-kt_{1/2}} = \frac{1}{2}$; applying ln gives $-kt_{1/2} = -\ln 2$, so $t_{1/2} = \ln 2/k$. A small k means a long half-life — this isotope is very long-lived.

Step 2: (b) 10% remaining

$$0.1 = e^{-0.005t} \implies t = \frac{\ln 10}{0.005} = 460.5 \text{ years}$$

"90% decayed" means 10% remains, so $N/N_0 = 0.1$. Apply ln: $-0.005t = \ln 0.1 = -\ln 10$, and the negatives cancel.

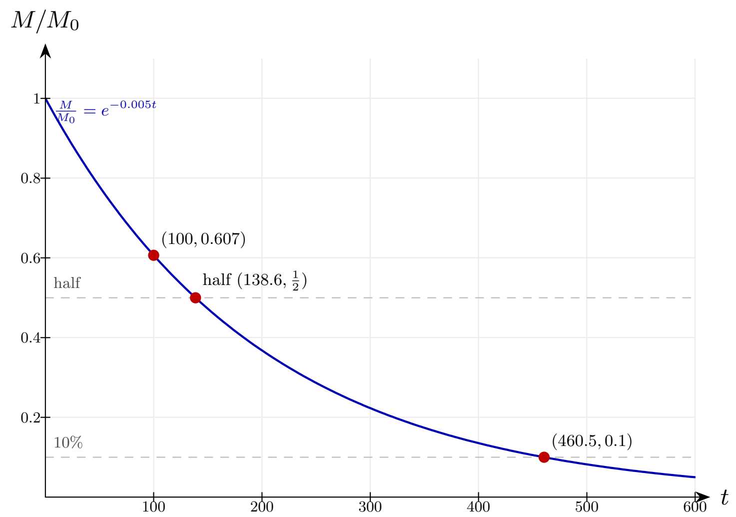
Step 3: (c) After 100 years

$$e^{-0.5} = 0.6065 = 60.7\%$$

At $t = 100$, the exponent is $-0.005 \times 100 = -0.5$, so the surviving fraction is $e^{-0.5} \approx 0.6065$ or about 60.7%. Less than one half-life has elapsed, so over half the substance is still present — consistent with $t_{1/2} \approx 139$ years from part (a).

Final answer: (a) 138.6 years (b) 460.5 years (c) 60.7%

Diagram.



The model plotted against time, with the key value(s) marked.